Python for Data Analysis

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1 Python for Data Analysis

These are the materials which are used for a course on *Python for Data Analysis* at the University of Freiburg in summer 2026. It does not assume any prior knowledge of python (although this might help). The goal of the course is some literacy in python, including basic knowledge such as data types and control flow, but also using libraries and APIs, reading and dealing with data, up to some methods such as PCA and regression.

Basically, the course has three parts: 1. Python basics: data types, functions, loops, conditions etc; (Lessons 01, 02, 03, 04) 2. Tools for data analysis: numpy, matplotlib, pandas; (Lessons 05, 05, 07, 08) 3: Project: choose data that interests you and creat a dashboard

Within the first two parts, you will be assigned homeworks, which you can find at the end of each section. For the third part, you will have to find a project by yourself.

Let us discuss some preliminairies.

Your python environment: We will assume that python and jupyter is installed on your system. These course notes are written with python 3.13, but other versions might work as well. In addition to plain `python`, we will need some libraries. Which ones we need (including their versions) is written in the file `requirements.txt`. Some jupyter notebooks might not work if these are not installed on your system. (To be more precise, they must be installed on the jupyter server which runs the notebools.)

General note: We avoid repeating the distinction between the commands `python` and `python3`, and will simply write `python3` in the sequel.

Starting a Jupyter notebook:

There are basically three ways `python` commands can be run: 1. In interactive mode on the command line by typing `python3` in your shell, 2. In `python`-scripts, usually ending with `myScript.py`. You can run these using `python3 myScript.py`. 3. Jupyter notebooks have the opportunity to mix usual text with `python` commands. We will use these in this course.

For Jupyter notebooks, there are at least two ways to run them:

1. Start the Jupyter server using `jupyter lab` (hopefully, it is already installed), which opens your browser where you can edit and save what you have done;
2. Use `vscode` as you IDE and work from there; see below.

Integrated Development Environment (IDE): When programming, we need to type code. This is possible in a code editor, and best done in a more sophisticated environment such as `vscode` or pycharm. It is best to acticvate your vertual environment using `source venv/bin/activate` before starting your IDE. In order to start vscode, go to the main folder of the course notes, and type `code .` E.g., vscode can be used for all three ways to write Python commands:

1. Use `Terminal -> New Terminal` in order to open a terminal, type `python3` and you are in interactive mode,
2. Write a Python script and run it using the opoened terminal,
3. Click on some Jupyther notebook, and vscode starts the Jupyter server for you.

A first Python cell: Here is the first cell of Python-code:

```
x = "Hello, "
y = "world!"
print(x+y)
```

```
Hello, world!
```

If you are working within the jupyter notebook (and not just in the html-file), you can execute this piece of code: Press the play bottom on the left (or hit `Control + Enter` within the Python-cell).

Using AI (ChatGPT, Gemini etc, Copilot, Claude, etc):

During the last years, AI has increasingly improved in code writing. You are encouraged to use this new technology! As usual when learning new things, this new tool can help a lot, and will be able to solve the exercises within the Jupyter notebooks. However, you still have to undestand the code, and what might go wrong during execution. As a first course in Python, you will end up having up to a few hundred lines of code, which has to be coherent. You must know the structure of that code, and in order to extend it, fix bugs etc. If you are ever involved in bigger projects, this is even more important.

For these course notes, I have also used AI (chatGPT, gemini) in order to ask the AI which topics I should cover, and make concrete suggestions for a workflow. (While I have been programming in Python3 for some years now, my experience with `numpy` and `matplotlib` was still limited when I started to write the course, since I have used `R` for such topics in the past.) Not all suggestions from the AI were great. However, I think that the end result for that course is much better than without the help of AI. I hope this also helps the learning experience.

2 Simple Data types

Download notebook.

2.1 Overview

Every variable or other object in python has a type. This is the kind of information which is stored insice the variable or object Here is an overview over the most important simple data types.

Type	Example	Description
`int`	`5`	Integer
`float`	`3.14`	Floating-point number
`bool`	`True`	Boolean value
`str`	`"hello"`	Text string
`NoneType`	`None`	No value / null value

- You can find about the type of any expression `x` in Python using `type(x)`.
- In fact, `type(x)` is already the first example of a built-in-function. The function `type` takes a single value as input, and returns its type. We will encounter various more functions already in this section. We will learn more about functions (built-in and self-written) in Section 3.4.
- There are conversions between some types. For example, `"0"` is a string, but `int("0")` (again a function, taking a single value, and returning its integer value) the integer `0`. Conversely, `str(0)` is the string `"0"`. We will have more examples below.

2.2 Integers (`int`)

With `int`s (numbers …,-2,-1,0,1,2,…), you can compute as usual.

```
x = 7
y = 2
print("x + y = ", x + y)
print("x * y =  ", x * y)
```

```
x + y =  9
x * y =   14
```

```
# x to the power of y
print("x ** y = ", x ** y)
```

```
x ** y =  49
```

```
# x divided by y (which gives a `float`)
print("x / y = ", x / y)
```

```
x / y =  3.5
```

```
# integer-valued division of x by y
print("x // y = ", x // y)
```

```
x // y =  3
```

```
# x modulo y
print("x % y = ", x % y)
```

```
x % y =  1
```

In most programming languages (e.g. `C++`), there is a minimal and a maximal possible value for an int. Python, however, does not have this limitation. Only the amount of available memory restricts the numbers which can be stored in an `int`.

2.3 Floating point numbers (`float`)

Generally speaking, floats are decimal numbers, which come with an exponent and a mantissa, e.g. $1.2 \cdot 10^2$. Python can do the same computations as with ints, but the following should be kept in mind:

- If you type `1`, Python thinks this is an int, but if you type `1.0` or `1.`, Python thinks this is a float.
- Unlike ints, floats do have a maximal and minimal element. The reason is that every float is stored in 64 bits, one for the sign, 11 for the exponent and 52 for the mantissa. As a result, floats range from about 10^{-324} to 10^{308}.

- When doing mixed computations with floats and int, e.g. `1.0 + 1`, the int is changed into a float.
- It is not possible to perform exact computations with some floats; see the example below.

```
x = 1.0
y = 1
print("x + y = ", x + y)
```

```
x + y =  2.0
```

```
print("0.1 + 0.2 = ", 0.1 + 0.2)
```

```
0.1 + 0.2 =  0.30000000000000004
```

Special floats are `inf`, `-inf` and `nan` (not a number). Let us see them in action:

```
print((10. ** 308) * 10)
```

```
inf
```

```
print((10 ** 308) * 10)
```

```
100000000000000
```

```
print(int(0.0))
```

```
0
```

```
print(int(0.5))
```

```
0
```

If you deliberately want to introduce `nan` or `inf`, you can do this:

```
print(float("inf"))
print(float("nan"))
```

```
inf
nan
```

On `float`, you can use many functions. We only name two at the moment, for some float `x`:

- `abs(x)`: The absolute value of `x`.
- `round(x)` and `round(x,n)`: Round `x` to the next `int`, or after `n` digits.

In Section 5.2, we will encounter that many more usual functions (like `sqrt`) can be used on floats, but we need to `import` the mathematical library for this.

2.4 True-False variables (`bool`)

There are exactly two boolean values, known as `True` and `False`. You can compute with them using `not`, `and`, `or` and 'xor'. In addition, `True` and `False` can be the result of a condition, such as the expression `1 == 2`.

```
x = True
y = False
print("not x =", not x)
```

```
not x = False
```

```python
print("x and y =", x and y)
```

```
x and y = False
```

```python
print("x or y =", x or y)
```

```
x or y = True
```

```python
print("x xor y =", x ^ y)
```

```
x xor y = True
```

```python
print("((not x) == y) =", (not x) == y)
```

```
((not x) == y) = True
```

```python
print("(1 == 2) =", 1 == 2)
```

```
(1 == 2) = False
```

```python
# Python converts `bool` into `int`
print("x + y =", x + y)
```

```
x + y = 1
```

```python
# != means "not equal"
print("(1 != 2) =", 1 != 2)
```

```
(1 != 2) = True
```

```python
# < refers to less than
print("(1 < 1) =", 1 < 1)
```

```
(1 < 1) = False
```

```
# <= refers to less than or equal
print("(1 <= 1) =", 1 <= 1)
```

```
(1 <= 1) = True
```

```
z = float("nan")
print("(z == z) =", z == z)
```

```
(z == z) = False
```

2.5 Text (`string`)

In many cases, strings are objects in quotation markes, such as in "'Hello, world!".

- **Operations on strings**: You can concencatenate strings using `+` as in "'Python" + "for" + "Data" + "Analysis" (but you cannot subtract strings). You can also repeat a string by multiplying it with an int.
- **f-strings**: They were introduced in Python 3.6. They are the way to go if you want to mix description and variables. They are written as `f"Hello, {name}"`. The `f` introduces the f-string. Within curly braces, there is a variable, which should be printed instead of the variable name.
- **Special characters**: There are many of them, we only mention `\t` (tabulator) and `\n` (new line) here.
- **Multiline strings**: Usually, python interprets a new line such that all commands are finished. However, if you want to use a sttring spanning multiple lines, you don't want this. Multiline strings start and end with `"""` (a triple quotation mark).
- **Quotation marks**: In various cases, you can use `'` instead of `"`. Sometimes this is useful when nesting quotation marks.
- **Substrings**: We will learn more about substrings, but here is already a way to see if a substring occurs within a string or not. The value of `"th" in "Python"` amount to `True`.

```
x = "Hello,"
y = "world"
# Concatenating two (or more) strings
print(x + y + "!")
```

```
Hello,world!
```

```python
# repeating a string
z = "bla"
print(z*3)
```

```
blablabla
```

```python
# a special character
print("\U0001F60E")
```

```python
# an example of an f-string
name = "Alice"
print(f"Hello {name}  ")
```

```
Hello Alice
```

```python
# a multiline string
text = """This is a multiline string.
It can span several lines.
Each line break is preserved."""
print(text)
```

```
This is a multiline string.
It can span several lines.
Each line break is preserved.
```

```python
# determine if a given substring occurs within a string
print("Is 'th' a substring of 'Python'? ", 'th' in 'Python')
print("Is 'the' a substring of 'Python'? ", 'the' in 'Python')
```

```
Is 'th' a substring of 'Python'?  True
Is 'the' a substring of 'Python'?  False
```

For strings, there are various build-in functions. We mention some of them. Here, `s` is a string.

- `str.strip(s)`: Deletes spaces in the front and back.

- `len(s)`: gives the length of the string.
- `str.replace(s, "x", "y")`: Replaces each occurrence of `"x"` in `s` by `"y"`.
- `str.split(s, "x")`: Splits `s` at each occurrence of `"x". The result is a list`‘, which we will encounter in Section 4.1. “,”.join
- `str.lower(s)`: Return the same string, but all letters are lowercase.
- `str.upper(s)`: Return the same string, but all letters are uppercase.
- `str.isalpha(s)`:
- `str.isdigit(s)`: Returns `True` if the string is actually a number, and `False` otherwise.
- `str.isalnum(s)`: Returns `True` if the string consists only of letters and numbers, and `False` otherwise.
- `str.isspace(s)`: Returns `True` if the string consists only empty spaces, and `False` otherwise.
- `str.islower(s)`: Returns `True` if the string consists only of lower case letters, and `False` otherwise.
- `str.isupper(s)`: Returns `True` if the string consists only of upper case letters, and `False` otherwise.
- `str.count(s, "x")`: Counts how ofter `"x"` appears in `s`.

Here is another very useful notation convention, known as dot-notation: For a function `str.something(s,...)`, where `s` is of type `str`, you can as well write `s.something()`. For example, `s.strip()` is the same as `str.strip(s)`.

2.6 Missing things (`NoneType`)

When analysing data, it is common to have missing values. These usually come as `None`. This is different from anything else we encountered so far (including `nan` and `False`).

```
x = None
print(x)
```

```
None
```

```
print("(x is None) =", x is None)
```

```
(x is None) = True
```

```
print("(None == False) =", None == False)
```

```
(None == False) = False
```

```python
print("(None == nan) =", None == float("nan"))
```

```
(None == nan) = False
```

2.7 Exercises

Exercise 1: Find out how to use scientific notation in Python. (If you don't know what this is, please find out!)

Exercise 2: Is there a Python package which can compute `0.1 + 0.2 == 0.3` without error?

Exercise 3: Print `"Hello, world!"` including the quotation marks in the output.

Exercise 4: Do newlines and tabs `\n`, `\t` count in `str.isspacce()`? How do numbers interact with `str.islower()`? If `s` is the strign of a number, what is `s.lower()`? Play around a bit with these fucntions and edge cases you might see in data.

Exercise 5: If you are on Windows, it is annoying that file paths come with `\`, which is interpreted as the start of a special character. Find out two ways how to give the string of the file path `C:\Users\Alice\Documents\file.txt`.

Exercise 6: What is `bool(None)`? What is `int(None)`?

Exercise 7: In tables, we often have `""` for a missing value. Is this the same as `None`?

3 Control flow – conditions, loops, functions, etc

Download notebook.

Coding starts to become interesting when we can control how variables change, and repeat operations.

3.1 Conditions (`if ... elif ... else`)

As a simple (mathematical) example, let us compute the minimum of two numbers., which works by an `if`-condition:

```
a = 2
b = 5
if a < b:
    res = a
else:
    res = b
print(f"The minimum of {a} and {b} is:", res)
```

```
The minimum of 2 and 5 is: 2
```

In general, the structure of an `if`-statement reads:

```
if _condition1_:
    case1
elif _condition2_:
    case2
elif _condition3_:
    case3
...
else:
    case_else
```

Here, read `elif` also `else if`, which means that the code jumps into this case if the respective condition is the first to hold. The `else` case only applied if both the `if` and all `elif` conditions do not apply.

In fact, there are abbreviations in python3 which are very handy. In particular, look at this piece of code.

```
res = a if a < b else b
print(f"The minimum of {a} and {b} is:", res)
```

```
The minimum of 2 and 5 is: 2
```

Let us make another example, also using `elif`. Given some `n`, the task is to print `Fizz` if `n` is a multiple of 3, `Buzz`, if it is a multiple of 5, and `FizzBuzz` if both conditions are satisfied.

```
n = 165
res = ""
if n % 3 == 0 and n % 5 == 0:
    res = "FizzBuzz"
elif n % 3 == 0:
    res = "Fizz"
elif n % 5 == 0:
    res = "Buzz"
print(res)
```

```
FizzBuzz
```

3.2 Loops with fixed length (`for`)

There are `for`-loops in Python3, as discussed in this section, and `while`-loops in the next section. The basic use-case of a `for`-loop is repeating a task several times. As a simple example, adding `b` a number of `a` times, results in `a * b`.

```
a = 3
b = 7
res = 0
for i in range(a):
    res = res + b
print(f"{a} * {b} = {res}")
```

```
3 * 7 = 21
```

Importantly, `range(a)` is a synonym for the numbers `0,..., a-1`. (In particular, note that Python3 usually starts counting at `0` and not at `1`, which will also be important in the next chapter.)

The general structure of a `for`-loop is as follows:

```
for _var_ in _iterable:
    _sequence of instructions_
```

3.2.1 Greatest common divisor I

Let us look at a more interesting example, computing the greatest common divisor (gcd) of two numbers, `a` and `b`. A very simple algorithm is going through all numbers `1,...,` in order, and update result each time we encounter a number which divides both, `a` and `b`. (Here, `range(1,a+1)` is a synoym for the numbers `1,...,a`.)

```python
# Basic algorithm for finding the gcd of two numbers
a = 105
b = 33

res = 1
for i in range(1,a+1):
    if a % i == 0 and b % i == 0:
        res = i
print(f"The gcd of {a} and {b} is {res}.")
```

```
The gcd of 105 and 33 is 3.
```

Let us make some remarks on `for`-loops: * In the last example, note that the instructios within the `for`-loop depend on `i`. * Within a `for`-loop, you can use `continue` in order to stop the execution of the current iteration, and immediately jump to the next `i`. * Within a `for`-loop, you can use `break` in order to stop the whole execution of the `for`-loop.

3.3 Loops with unknown length (`while`)

The `for`-loop directly iterates over some variable (`i` above). The `while`-loop instead comes with a condition, which is checked everytime it is started new. The general structure is:

```
while _condition_:
    instructions which might change the value of _condition_
```

Let us make two examples: finding $\sqrt{2}$ and computing the gcd using Euclid's algorithm.

3.3.1 Finding $\sqrt{2}$

For finding $\sqrt{2}$, note that this is a fixed point for the iteration

$$x_{n+1} = \frac{x_n}{2} - \frac{1}{x_n}.$$

(In order to see this, assume that $x = x_n = x_{n+1}$, and multiply the recursion by x.) We can use this as follows.

```python
x = 1
eps = 1e-10
x_prev = 0

while abs(x - x_prev) > eps:
    x_prev = x
    x = x/2 + 1 / x
print(x)
```

```
1.414213562373095
```

3.3.2 Finding the greatest common divisor II

The above algorithm (using `for`-loops) for finding the gcd were not particulary efficient. (You might want to test them with some large numbers.) Euclid's algorithm is much more efficient. It is based on the observation that `c % a == 0` (`c` divides `a`) and `c % b == 0` iff `c % a == 0` and `c % (b % a)`.

In order to see this, note that `c % a == 0` and `c % b == 0` iff `c % a == 0` and `c % (b - da) == 0` for any `d`. Then, the result follows from choosing `d = b \\ a` since `b - (b \\ a) * a = b % a`. Moreover, every number divides `0`, so if `b % a == 0`, the gcd of `a` and `b` is `a`.

We can use these insights, taking `a < b` for simplicity. Starting with `a` and `b`, we compute if `b % a == 0`. If yes, we are done, and return `a`. If not, we compute the gcd of `a` and `b % a` instead.

```
# Euclid's algorithm for finding the gcd of two numbers
a_input = 105
b_input = 33
# Copy the input to new variables
a = a_input
b = b_input
# Run Euclid's algorithm
while a != 0:
    r = b % a
    print(f"b = {b}, a = {a}, remainder = {r}")
    b = a
    a = r
print(f"The gcd of {a_input} and {b_input} is {b}.")
```

```
b = 33, a = 105, remainder = 33
b = 105, a = 33, remainder = 6
b = 33, a = 6, remainder = 3
b = 6, a = 3, remainder = 0
The gcd of 105 and 33 is 3.
```

3.4 Functions (`def`)

Once we implement something, we wan to reuse it without re-writing the code. This means, we want to pack our code into its own function for reusing it. For Euclid's algorithm, this would look as follows:

```
a = 105
b = 33

def gcd(a,b):
    """ Compute the greatest common divisor of a and b."""
    while a != 0:
        r = b % a
        b = a
        a = r
    return b
```

```python
print(f"The gcd of {a} and {b} is {gcd(a,b)}.")
```

```
The gcd of 105 and 33 is 3.
```

The general structure is

```
def name_of_the_funcion(arg1, arg2, arg3 = default3, arg4=default4):
    some instructions
    return something
```

Not all functions return something. Those who don't usually produce some output. Here is an example

```python
def greeting(name, prefix="Hello"):
    """ Return a greeting for name."""
    print(f"{prefix}, {name}!")

greeting("Alice")
greeting("Bob", prefix="Hi")
greeting(prefix="Welcome", name="Charlie")
```

```
Hello, Alice!
Hi, Bob!
Welcome, Charlie!
```

Here are some important remarks on functions:

- Variables within functions are private to them. (There will be exceptions to this) In the gcd example, variable names `a` and `b` occur both, within and outside of the function. Although the variables `a` and `b`, which are private to the function, are changed within the function, the `print`-statement still knowns their value from before the function definition.
- Variables which are given to functions might have default values, as the `prefix = "Hello"` in the `greeting` example.
- For giving variables, there are two options:
  - you can rely on the position they have in the definition of the function. (Example: `greeting("Bob", prefix="Hi")`, where you rely on `"Bob"` being in the `name` position.) These are called positional variables.

  - you can give it the explicite variable name it has in the function definition. In this case, the order of variable names does not matter. (Example: In `greeting(prefix="Welcome", name="Charlie")`, the output was correct although the order of the two variables differs from the function definition.)

- All positional arguments must come before keyword aguments.
- Both functions from above have a *docstring*, which is the multiline string after the `def`-line. They indicate what this function is doing and are displayed in various places, e.g. when you hover over a function in your jupyter notebook.
- Sometimes, it is too much to define an own function for a very simple task. In this case, Python3 comes with `lambda`-functions, which are very quick: `lambda x: x * x`. Most use cases are with container data types, and we will have one in Section 4.1.

3.5 Exception handling (`raise`, `try ... except`)

In code, various things happen, and some go wrong. Raising (throwing) errors, catching them, and dealing with them is the topic of this section. There are many error types implemented. Let us look at some of them:

Exception	When it happens	Example
`ValueError`	Right type, wrong value	`int("abc")`
`TypeError`	Wrong type used	`"2" + 3`
`IndexError`	Index out of range	`[1,2][5]`
`KeyError`	Dictionary key missing	`{"a":1}["b"]`
`ZeroDivisionError`	Division by zero	`1 / 0`
`FileNotFoundError`	File does not exist	`open("x.txt")`
`PermissionError`	No access rights	opening protected file
`AttributeError`	Object has no attribute	`"abc".foo`
`NameError`	Variable not defined	`print(x)`
`ImportError`	Import fails	`import nonexisting`
`ModuleNotFoundError`	Module not found	`import xyz`
`AssertionError`	`assert` fails	`assert False`

We obtain an error here:

```
print("This will not work")
print(f"1/0 = {1/0}")
print("If this print works, the program has continued past 1/0")
```

```
This will not work
```

```
ZeroDivisionError: division by zero
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
Cell In[10], line 2
      1 print("This will not work")
----> 2 print(f"1/0 = {1/0}")
      3 print("If this print works, the program has continued past 1/0")
ZeroDivisionError: division by zero
```

However, if we *catch* the error, the code can run past it:

```python
try:
    print("This will not work")
    print(f"1/0 = {1/0}")
    print("This never runs")
except ZeroDivisionError as e:
    print("You must not divide by zero!")
    print("This is the error message:", e)
print("The program continues!")
```

```
This will not work
You must not divide by zero!
This is the error message: division by zero
The program continues!
```

Now we know how to *catch* errors, but often, we need to *throw* them. This is actually simple using `raise`:

```python
def compute_sth(x):
    if type(x) != int:
        raise TypeError("x must be an int!")
    y = x ** x
    return y

try:
    print(compute_sth(2))
    print(compute_sth("abc"))
except TypeError:
    print("Calculation not possible, sorry")
```

```
4
Calculation not possible, sorry
```

3.6 Context management (`with`)

Often, our Python3 file (or notebook) must interact with other files or ressources. In order to do this safely, we are using `with`.

xxx file handling!

3.7 Exercises

Exercise 1 Can you code the `FizzBuzz` example in a single line of code? So, dependent on some `n`, give one line of code, which gives `Fizz` if `n` is a multiple of 3, `Buzz`, if it is a multiple of 5, and `FizzBuzz` if both conditions are satisfied.

Exercise 2 Multiple `elif`s in a row are not very elegant. Python3 (since version 3.10) comes with `match`. Find out how you can change the `FizzBuzz` example of the main text by using `match` instead of `elif`s.

Exercise 3 Write a version of the code for finding the gcd by running (not from 1 to `a`, but) from `a` to `1`, and see if the corresponding number divides both, `a` and `b`. Apparently, you can break the `for`-loop when you found the first such number. Implement that breaking of the `for`-loop.

Exercise 4 Compute $\sum_{i=1}^{100} i$ using a `for`-loop.

Exercise 5 Find the largest n such that $\sum_{i=1}^{n} i < 1000$.

Exercise 6 `for`-loops can also become nested. Compute $\sum_{i=1}^{100} \sum_{j=1}^{i} j$. (Note the double indentation for the inner `for`-loop!)

Exercise 7 Define a function computing the sum of the first `n` integers.

Exercise 8 Write a function (without using imports) which computes the sum of all digits (cross sum) of an int.

```
def check_age(age_as_string):
    try:
        age = int(age_as_string)
    except ValueError as e:
        print("Age must be an int.")
        age = -1
    return age

print(check_age(25))
print(check_age("abc"))
```

```
25
Age must be an int.
-1
```

4 Container data types

Download notebook.

When working with data, we usually not only have *one* int or *one* str, but many of them. For this situation, there are the container data types, `list`, `tuple`, `set`, and `dict`.

4.1 Lists (`list`)

The notation for a list comes with square brackets, where the entries are separated by a comma. The list of numbers 1, 2, 3 is `[1,2,3]`. A list of names is e.g. `["Alice", "Bob", "Charlie", "Donna"]`. So, for a `list`, the entries have simple (or again container) data types.

- We can access the entries of a list using square brackets. However, we have to be careful if the length of the list is too short.
- We can access sublists by using `i:j` for the entries `i,...,j-1`. If we want to have the beginning until `j-1`, we use `:j`. If we want everything after `i`, we use `i:`.
- Elements of a list can be lists (or other container data types) themselves.
- The types of elements of a list can be mixed.
- When talking about `for`-loops, we already had `range(n)`, which stands for `0,...,n-1`. This is not exactly a list, but can be turned into one by `list(range(n))`. (More precisely, there is also `range(start, stop, step)` for `start, start + step, start + 2*step,...`, which stops at `stop-1` at the latest.)

```
# a list of ints
x = [1,2,3]
print(x[0])
print(x[3] if len(x) > 3 else None)
```

```
1
None
```

Slicing refers to sublists.

```python
x = [1,2,3]
print(f"The first and second element: {x[0:2]}")
print(f"The first two elements: {x[:2]}")
print(f"All elements after the first element: {x[1:]}")
```

```
The first and second element: [1, 2]
The first two elements: [1, 2]
All elements after the first element: [2, 3]
```

Playing around with `range`.

```python
print(f"The first and second odd number: {list(range(1, 5, 2))}")
print(f"The first and second odd number: {list(range(1, 100, 2))[:2]}")
```

```
The first and second odd number: [1, 3]
The first and second odd number: [1, 3]
```

The elements of a list can be lists themselves.

```python
y = [[1,2], [3,4,5], 6, 7, 8]
print(y[1])
print(y[1][0])
```

```
[3, 4, 5]
3
```

A list can have mixed types.

```python
z = [1, "two", [3,4], 5.0]
print(z[1])
```

```
two
```

Here are some functions on lists.

- `len(l)`: length of `l`.
- `sum(l)`: sums all elements of `l`, if possible.
- `min(l)`, `max(l)`: computes the minimum/maximum of `l`.
- `sorted(l)`: returns a sorted version of `l`.

- `filter(fun, l)`: returns a filtered list, which only contains elements `x` of `l` with `fun x == True`.
- `list.append(l, x)` (or `l.append(x)`) (where `x` is any object) appends the element `x` to the list `l`.
- `list.extend(l, x)` (or `l.extend(x)`) (where `x` is a list) extends `l` by all elements of `x`. (You can also write `l + x` for this.)
- `list.insert(l, i, x)` (or `l.insert(i,x)`) inserts `x` at position `min(i, len(l))` in `l`.
- `list.remove(l, x)` (or `l.remove(x)`) removes the first occurrence of `x` in `l`.
- `list.pop(l)` (or `l.pop()`) removes the last element from `l`.
- `list.index(l, x)` (or `l.index(x)`) gives the index of the first occurrence of `x` in `l`.
- `list.count(x)` (or `l.count(x)`) counts the number of occurrences of `x` in `l`.
- `list.sort(l)` (or `l.sort()`) sorts the list.
- `list.reverse(l)` (or `l.reverse()`) reverses the list.

For the functions starting with `list`, there is a catch: While we mentioned that a function takes its input (here the list), and does not alter it, `list`s are mutable objects, which means that they are changed when calling such a function on them. So, e.g. after `l.sort()`, the list `l` itself is sorted, and there is no return value of that function (more precisely, the return value is `None`.) This can be counterintuitive:

```
l = [3,2,1]
# This gives none since the return value of l.sort() is None
print(l.sort())

l = [3,2,1]
# This changes l and makes it sorted.
l.sort()
# Now we can print it
print(f"l = {l}")

# In contrast, sorted() does not change the list but returns a new sorted list.
l = [3,2,1]
print(sorted(l))
```

```
None
l = [1, 2, 3]
[1, 2, 3]
```

Assume you want to sort a `list` not as usual. In this situation, you will find out about the `key`-parameter in `sorted`. It takes a function, we sorts the list according to the values of this function. As an example, assume we want to sort a list of `int`s according to their abolute value:

```
l = [5, -4, 0, 7, -3, 2]
print(sorted(l, key = lambda x: abs(x)))
```

```
[0, 2, -3, -4, 5, 7]
```

Similarly, you can e.g. filter the list by only taking numbers which are at least one:

```
l = [5, -4, 0, 7, -3, 2]
print(list(filter(lambda x: x >= 1, l)))
```

```
[5, 7, 2]
```

Let us shortly describe two functions on a list of `bools` `l`.

- `all(l)` gives `True` if all entries of `l` are 'True.
- `any(l)` gives `True` if at least one entry of `l` is 'True.

4.2 Tuples (`tuple`)

At first glance, a `tuple` is like a `list`, but comes with normal brackets `()` instead of square brackets `[]`. So, `(1,2,3)` is the tuple of these three numbers, and `("Alice", "Bob", "Charlie") is a tuple of three names. For the tuplel, still,`l[0]`gives the first entry etc. The main difference between`tuple`and`list`is that`lists are more flexible when adding or changing entries, and`tuple's are fixed (immutable) once they have been created.

```
l = ("Alice", "Bob", "Charlie")
# The next line does not work, since l is immutable
# l[0] = "Anna"
l = ["Alice", "Bob", "Charlie"]
# The next line works, since l is mutable.
l[0] = "Anna"
print(f"l = {l}")
```

```
l = ['Anna', 'Bob', 'Charlie']
```

An important aspect of tuples is that single values can easily be packed into a tuple. An important example is to use a tuple in order to assign two or more values at the same time:

```python
def min_max(a,b):
    (mi, ma) = (a if a < b else b, b if a < b else a)
    return mi, ma

a = 20
b = 10
res = min_max(a,b)
print(f"Minimum and maximum are {res[0]} and {res[1]}.")
```

```
Minimum and maximum are 10 and 20.
```

A special note applies to tuples of length 1. Here, note that e.g. `(1)` does not create a `tuple` of length 1, but an `int`. Instead, use `(1,)` for the `tuple` of length one.

```python
thisisnotuple = (1)
print(type(thisisnotuple))
print(thisisnotuple[0])
thisisatuple = (1,)
print(type(thisisatuple))
print(f"Its first and only element is {thisisatuple[0]}.")
```

```
<class 'int'>

TypeError: 'int' object is not subscriptable
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[11], line 3
      1 thisisnotuple = (1)
      2 print(type(thisisnotuple))
----> 3 print(thisisnotuple[0])
      4 thisisatuple = (1,)
      5 print(type(thisisatuple))
TypeError: 'int' object is not subscriptable
```

4.3 Sets (`set`)

As in mathematics, sets have no special order, and there are noduplicates within a set. In Python3, they can be created using curly brackets, so e.g. `{1,2,3}` as well as `{1,3,2,3,1}` is the set of the numbers `1,2,3`. Although there are functions for updating sets (as for lists), the

most frequent application is for removing duplicates. For example, if you have a `list`, which can contain duplicates, `list(set(l))` is the same list (maybe the order changed) but with duplicates removed.

Here are some functions on some `s` and `t`: * `len(s)`: returns the number of unique elements in `s`. * `s | t`: The union of `s` and `t`. * `s & t`: The intersection of `s` and `t`. * `s - t`: The difference of `s` and `t`.

```
l = [1,3,2,4,3,2]
print(f"There are {len(set(l))} unique elements in {l}.")
```

Unions, intersection, and set differences work as expected:

```
s = {1,2,3}
t = {5,4,3}
print(f"The union of {s} and {t} is {s | t}.")
print(f"The intersection of {s} and {t} is {s & t}.")
print(f"The difference of {s} and {t} is {s - t}.")
```

4.4 Dictioanries (`dict`)

Dictionaries (`dict`) are nothing but a collection of key-value pairs. Here is an example:

```
d = {
        "first_name": "Alice",
        "last_name": "Müller",
        "age": 29,
        "hobbies": ["reading", "cycling", "chess"]
    }
print(f"""{d['first_name']} {d['last_name']} is {d['age']} and likes
    {', '.join(d['hobbies'])}.""")
```

```
Alice Müller is 29 and likes
    reading, cycling, chess.
```

- As you see in the example, the f-string uses nested quotation marks, `"...'...'..."`. For this reason, we use single quotation marks `''` as inner separators and double quotation marks `""` for the outer quotation marks. If there is no nesting, there is no difference in using single/double quotation marks.
- keys in a dict can be any immutable data type (like `int`, `str`, `float`, `tuple`).

As you see, accessing field `'age'` in the `dict p` is done using `p['age']`. However, assume `'age'` is not a field in `p`. Then, `p['age']` gives an error. For this reason, it is often better to use the `get`-function:

```
print(f"{d['first_name']} {d['last_name']} lives in {d.get('address')}.")
# This version uses a default field in get
print(f"""{d['first_name']} {d['last_name']} lives
    in {d.get('address', 'unknown')}.""")
# We can also add "address" to the dict
d["address"] = "Freiburg, Germany"
print(f"""{d['first_name']} {d['last_name']} lives
    in {d.get('address', 'unknown')}.""")
```

```
Alice Müller lives in None.
Alice Müller lives
    in unknown.
Alice Müller lives
    in Freiburg, Germany.
```

Some of the most important functions on `d` as a `dicts` are: * `dict.keys(d)` (or `d.keys()`): Returns the list of all keys. * `dict.values(d)` (or `d.values()`): Returns the list of all values. * `dict.items(d)` (or `d.items()`): Returns the list of tuples (key, value).

```
print(f"The keys are {', '.join(d.keys())}.")
print(f"The values are:")
for v in d.values():
    print(str(v))
print(f"The key-value-pairs are:")
for key, value in d.items():
    print((key, str(value)))
```

```
The keys are first_name, last_name, age, hobbies, address.
The values are:
Alice
Müller
29
['reading', 'cycling', 'chess']
Freiburg, Germany
The key-value-pairs are:
('first_name', 'Alice')
('last_name', 'Müller')
```

```
('age', '29')
('hobbies', "['reading', 'cycling', 'chess']")
('address', 'Freiburg, Germany')
```

Let us mention some more functions on `dicts`:

- `dict.update(d, d')` (or `d.update(d')`): extends `d` by the keys in `d'`, and overwrites entries in `d` by the values of `d'`. Another possibility is `d |= d'` (similar to a set-union).
- `dict.pop(d, key, default)` (or `d.pop(key, default)`): removes the key `key` from `d` and returs its value (or `default`, if the key isn't found).
- `del d[key]`: Deletes the entry with key `key` from `d`.

```python
d |= {
    "profession" : "student",
    "subject" : "Mathematics"
}
del d["age"]
for key, value in d.items():
    print((key, str(value)))
```

```
('first_name', 'Alice')
('last_name', 'Müller')
('hobbies', "['reading', 'cycling', 'chess']")
('address', 'Freiburg, Germany')
('profession', 'student')
('subject', 'Mathematics')
```

Here is a last example for a list of dicts.

```python
people = [
    {
        "first_name": "Alice",
        "last_name": "Müller",
        "age": 29,
        "hobbies": ["reading", "cycling", "chess"]
    },
    {
        "first_name": "Bob",
        "last_name": "Schmidt",
        "age": 35,
        "hobbies": ["hiking", "photography", "cooking"]
```

```
    },
    {
        "first_name": "Clara",
        "last_name": "Weber",
        "age": 22,
        "hobbies": ["painting", "piano", "running"]
    },
    {
        "first_name": "David",
        "last_name": "Fischer",
        "age": 41,
        "hobbies": ["gardening", "woodworking", "cycling"]
    }
]
for p in people:
    print(f"{p["first_name"]} {p["last_name"]} is {p["age"]} and likes "
        + ", ".join(p["hobbies"]) + ".")
```

```
Alice Müller is 29 and likes reading, cycling, chess.
Bob Schmidt is 35 and likes hiking, photography, cooking.
Clara Weber is 22 and likes painting, piano, running.
David Fischer is 41 and likes gardening, woodworking, cycling.
```

4.5 Comprehensions

Python3 has very expressive (i.e. short) ways to alter container data types. The general form is (where "iterable" is a container data type):

```
[expression for variable in list if condition]
[expression_if_true if condition else expression_if_false for variable in list]
{key: value for item in dict if condition}
```

Let us make some examples, and you will see how powerful these complehensions are!

```
# The squares of all even numbers below 10
print([x*x for x in range(10) if x % 2 == 0])
# The names of people who like cycling
print("People who like cycling are " +
    ', '.join([f'{p["first_name"]} {p["last_name"]}' for p in people
        if "cycling" in p['hobbies']]) + ".")
```

```
[0, 4, 16, 36, 64]
People who like cycling are Alice Müller, David Fischer.
```

4.6 Exercises

Exercise 1 For two strings `s` and `t`, write a function `myCount(s, t)`, which counts the numbers of occurrences of `t` in `s`. Unlike `count`, it should count the number of positions in `s` starting with `t`. For example, `"bababab".count("bab")` gives 2, but `"bababab".myCount("bab")` should give 3.

Exercise 2 Find out how you can sort a list descendingly.

Exercise 3 Write a function which removes duplicates from a list (you might want to use `set` here), but keeps the order of the list.

Exercise 4 Given a list of names [“ALICE”, “BOB”, “CHARLIE”], convert them all to lowercase.

Exercise 5 From a list of fruits [“apple”, “banana”, “kiwi”, “mango”, “pear”], create a list of fruits that have exactly 5 letters.

Exercise 6 Create a list of tuples representing all possible `(x, y)` coordinates where `x` is from `[1, 2, 3]` and `y` is from `[4, 5]`.

Exercise 7 Given the dict `{"a": 1, "b": 2, "c": 3}`, create a new dictionary that swaps keys and values `{1: "a", ...}`. This is best done using a comprehension.

Exercise 8 Assume you have `a = [10, 20, 30, 40]` and `mask = [True, False, True, False]`, you might want obtain a sublist of `a` which only contains the positions where `mask` has `True. Find out (using AI) how this works (without using other libraries), and describe what your code does. (You will probably find out about the`zip‘-function, which you should look at more closely.)

5 Some useful libraries/packages/modules

Download notebook.

In this chapter, we take a little break. Although there are lots of things to learn, the material covered here has the role of helping to learn about the Python3 ecosystem and what is possible, rather than learn the details of various labraries. The first section, however, is central and will be needed in any Python3 project you will start.

5.1 Loading modules (`import`)

As in all programming languages, not all python programs come in a single file. If a Python3 project spans multiple files, an `import` becomes necessary. In Python3, there is syntax for `import`s as follows:

Syntax	Meaning
`import module`	full module
`import module as name`	alias
`from module import x`	specific objects
`from module import *`	everything (avoid)
`import module.submodule`	submodule

Generally, there are packages (folders), modules (a `.py`-file within a package, or a single `py`-file), and submodules. Every submodule is a module. In any case, the import loads the (sub-)module, and executes all top-level code inside it (i.e. everything which is not inside a function).

There are three cases:

1. You wrote some code in file `one.py` (e.g. some functions) which you want to use in file `two.py`. (Note, btw, that the first digit in `one.py` must be a letter.) Assume `one.py` is as follows:

```
# File: one.py

def myConst():
    return 42
```

Then, the import in `two.py` is either `import one` (both files must be in the same folder), or `from one import myConst` or (not recommended) `from one import *`. In the first case, you must prefix `myconst()` by `one`:

```
# File: two.py

import one
print(f"My constant is {one.myConst()}.")
```

In the second case, there is no prefix:

```
from one import myConst
# or
from one import *
print(f"My constant is {myConst()}.")
```

Observe that the latter option comes with the disadvantage that you cannot see from the call of `myconst()`, if it is defined in `two.py` or in`one.py`, so it becomes harder to keep track of your code if things become more complicated.

If you have a folder `myPackage`, which holds `one.py`, you can must write `import myPackage`, `import myPackage.one` or `import myPackage.one as myNameOne` with differing ways to use `myConst()`:

```
import myPackage
print(f"My constant is {myPackage.one.myConst()}.")

import myPackage.one
print(f"My constant is {one.myConst()}.")

import myPackage.one as myName
print(f"My constant is {myName.myConst()}.")
```

2. You want to use functions from a built-in library. Then, a simple `import module` is often the best choice. We will see examples in Section 5.2.
3. You want to use functions from a library, which needs to be installed. Here, before the `import module` even works, you need to install the module in your python environment. In a terminal, within your Python3 environment, use

```
# Uncomment the next line and run the cell for installation.
# This only needs to be done once for every package.
# If pip3 does not work, use pip instead.
# !pip3 install numpy
```

and then, `import numpy` will work. We will see examples for this e.g. in **?@sec-loading**.

5.2 Mathematics (`math`)

The mathematical library in Python is used by `import math`. Then, you have e.g. the following functions, which are self-explanatory: `math.sqrt()`, `math.exp()`, `math.log()`, `math.sin()`, `math.cos()`, and constants like `math.pi` and `math.e`.

5.3 Dates and times of day (`datetime`)

In data science, we often have to handle dates. Their disadvantage is that there is no standard way to write these as strings. In Python3, this and various other problems around dates and times is solved by the `datetime`-library.

Submodules (in fact, these are `class`es in this case) of this package are

- `date`: handling dates
- `time`: handling times
- `datetime.datetime`: handling `datetimes`, i.e. dates and times together
- `timedelta`: time differences
- `tzinfo` and `timezone`: handling timezones

We will here use

```
import datetime
```

at the cost that `datetime`-objects are only accessible by saying `datetime.datetime` (the first `datetime` being the package`, the second`datetime‘ being the class/submodule).

Let us create a `datetime`-object‘:

```
dt = datetime.datetime(2026, 4, 21, 12, 15)
print(dt)
print(f"year: {dt.year}")
print(f"month: {dt.month}")
print(f"day: {dt.day}")
print(f"hour: {dt.hour}")
print(f"minutes: {dt.minute}")
print(f"second: {dt.second}")
print(f"microsecond: {dt.microsecond}")
print(f"date: {dt.date()}")
print(f"time: {dt.time()}")
```

```
2026-04-21 12:15:00
year: 2026
month: 4
day: 21
hour: 12
minutes: 15
second: 0
microsecond: 0
date: 2026-04-21
time: 12:15:00
```

which is the starting time of the first session in this course. As we see here, we can extract year,..., microseconds from the `datetime`-object.

Often, we are given strings such as `"21.4.26, 12:15"`, and want to convert this into a `datetime` object. For this, we have `datetime.datetime.strptime`:

```
s = "21.4.26, 12:15"
dt = datetime.datetime.strptime(s, "%y.%m.%d, %H:%M")
print(s + f" is converted to {dt}.")
```

```
21.4.26, 12:15 is converted to 2021-04-26 12:15:00.
```

(For the correct way to place %, y, m, d etc., it is in fact best to consult your favourite AI.)

Reversely, we want to print a `datetime`-object in a certain way. For this, we have `datetime.datetime.strftime`:

```
print(dt.strftime("%-m/%-d/%y, %H:%M"))
```

```
4/26/21, 12:15
```

which is the US-style output.

Next, assume we want to compute the same time 12 weeks from now. For this, we use `datetime.timedelta`.

```
print(dt + datetime.timedelta(days=84))
```

```
2021-07-19 12:15:00
```

Here are some more remarks:

- `datetime.date` is not used very often. The reason is that you can recover a `date` from a `datetime`, but not the other way round.
- `datetime.time` has the same limitations.
- We will not cover time-zones here, but if you have global data, they will become important!

5.4 Decimal numbers (`decimal`)

We already saw that:

```
print(f"Rounding error: 0.2 + 0.1 = {0.2 + 0.1}.")
```

```
Rounding error: 0.2 + 0.1 = 0.30000000000000004.
```

The reason is that computers in general use a base of 2 for all computations, and 0.3 has a bad representation in this system. While this small error is acceptable often, it is not in some contexts (above all, finance). In this case, we can use the `decimal` library. Within `decimal`, a base of 10 is used, but calculations are slower than for `floats`.

```
from decimal import Decimal
print(f"Exact calculation: 0.2 + 0.1 = {Decimal('0.2') + Decimal('0.1')}.")
```

```
Exact calculation: 0.2 + 0.1 = 0.3.
```

5.5 Regular expressions (`re`)

Sometimes, you need to find patterns in a string. In data science, this happens if some field contains unstructured text, and you want to extract information. As an example, assume you load data from a csv-file, and one columns contains all email-adresses of the corresponding item. If you read this, you have something like `"muller@gmail.com; muller@yahoo.com"`. However, you want to separate the two addresses, and store them in a `list`. One way to approach this is using a regular expression (which in this case reads `"\S+@\S+\.\S+"`):

```
import re

text = "muller@gmail.com; muller@yahoo.com"

emails = re.findall(r"\S+@\S+\.\S+", text)
print(emails)
```

```
['muller@gmail.com;', 'muller@yahoo.com']
```

Let us quickly break this down to some extent, looking at `"\S+@\S+\.\S+"`: * `\S`: any non-whitespace character * `+`: one or more of the previous thing * `@`: exactly the `@` symbol * `\.`: exactly the dot `.` character (separating e.g. gmail from com)

We can agree that this exactly what an email address looks like. The `re.finall()` finds all places in `text` which match this pattern and extracts them, putting them into a list. Note that there are more patterns to find than `\S`, `+` and exact matches. For this reason, it is advised to use AI when it comes to regular expressions.

5.6 Storing and restoring objects (`pickle`)

Sometimes, you want to store intermediate objects. One way is to write them out in files, but you can also use the `pickle`-library in order to write Python3 objects to disc. Here is an example:

```
import pickle

data = {"name": "Alice", "age": 30}

# Save to file (pickle)
with open("data.pkl", "wb") as f:
    pickle.dump(data, f)
```

```
# Load from file (unpickle)
with open("data.pkl", "rb") as f:
    loaded_data = pickle.load(f)

print(loaded_data)
```

```
{'name': 'Alice', 'age': 30}
```

5.7 Testing code (pytest)

```
# Uncomment the next line and run the cell for installation.
# If pip3 does not work, use pip instead.
# !pip3 install pytest
```

When starting to code, producing reliable programs is both important and difficult. One – not the best – way to approach this is to rely on the program until something breaks. It is actually better to *test* the code, i.e. to write tests. This means that you tell your program that you expect a certain behavior.

A simple and powerful library for this is `pytest`. When working with `.py`-files, you usually separate such tests in a separate file. Every test-case is within a function with name starting with `test_`. In a terminal, you can run `pytest` on that file, which performs all tests, and reports any error.

However, we are working with Jupyter notebooks here, so we cannot run `pytest` on separate files. Although there are workarounds (the package `ipytest`), we will only go through some basics here. This means we are introducing `assert`.

```
x = 5
y = -1
assert 2 + 2 = 4
assert 2 + 2 = 5
```

```
SyntaxError: invalid syntax (2660193817.py, line 3)
  Cell In[12], line 3
    assert 2 + 2 = 4
                 ^
SyntaxError: invalid syntax
```

5.8 Using the operating system (`os`)

5.9 Interacting with the environment (`sys`)

xxx command line arguments

5.10 Exercises

Exercise 1 What is your age in seconds?

Exercise with datetime

date,sales 2026-01-01,120 2026-01-02,135 2026-01-03,128 2026-01-08,150 2026-02-01,210 2026-02-03,215

We want to:

load the data

convert the date column to datetime

extract year/month/weekday

filter by date

compute time differences

resample by month

create rolling averages

Exercise f Apparently, `"muller@gmail.com; wrong-email; muller@yahoo.com"` only contains two valid email addresses. Use AI and `re` in order to produce and understand code, which separates the correct from the incorrect email addresses.

Exercise d A bank manager has an idea: Interests should be computed daily (as usual), but rounded to cents (round up if last digit is 5 or above) at the end of the day. (So, sub-cents are not kept for computing the account balance.) Assume someone has invested 1 Euro on January 1st, 0 (birth of christ) with an interest rate of 5% per year. * Compute the account balance *as usual*, i.e. keeping sub-cents at all days. * Compute the account balance using the above strategy, i.e. ignoring sub-cents at the end of each day.

6 Numerical mathematics

Download notebook.

We will be using numpy {#sec-numpy} and scipy

6.1 Linear algebra (`np.array` etc)

This package provides powerful and fast computations for numerical mathematics. (Most of it is implemented in `C`.) We will only learn about some parts of it, some linear algebra and some optimization. Usually, it is imported via

```
import numpy as np
```

(Then, the functions from `numpy` are called `np.something`.) Together with `pandas` (see Section 7.3) for data handling, this is the basic framework for data analysis in Python3.

Let us start with linear algebra. A vector or matrix is an `np.array`. `numpy` interprets a vector as a column vector, and

```
x = np.array([1.0, 2.0, 3.0])
A = np.array([
    [1.0, 0.0, 2.0],
    [0.0, 1.0, 1.0]
])

print("x =")
print(x)
print()
print("A =")
print(A)
print()
print("x.shape =", x.shape)
print("A.shape =", A.shape)
print("x.ndim =", x.ndim)
```

```
print("A.ndim =", A.ndim)
print("x.dtype =", x.dtype)
```

```
x =
[1. 2. 3.]

A =
[[1. 0. 2.]
 [0. 1. 1.]]

x.shape = (3,)
A.shape = (2, 3)
x.ndim = 1
A.ndim = 2
x.dtype = float64
```

Generally we

6.2 Statistics (`scipy.stats`)

The `scipy`-library is a powerful which covers much of numerical mathematics. It comes with several submodules, e.g. `scipy.linalg` (linear algebra), `scipy.optimize` (optimization), `scipy.integrate` (numerical integration and ODE solving), and `scipy.signal` (signal processing). It works best with `numpy`-arrays, which we will cover in xxx.

Here, we only describe what `scipy` has to offer in descriptive statistics. The function names are mostly self-explanatory:

from scipy import stats

stats.norm.mean() # mean = 0 stats.norm.pdf(0) # density at 0 stats.norm.cdf(1.96) # cumulative probability

data = [1, 2, 3, 4, 5]

stats.describe(data)

stats.ttest_1samp(data, popmean=3)

stats.ttest_ind(sample1, sample2)

stats.chisquare([10, 20, 30])

stats.pearsonr(x, y)

stats.scoreatpercentile(data, 50) # median

stats.norm.rvs(size=5)

stats.tmean(data) # mean stats.tstd(data) # standard deviation stats.tvar(data) # variance

6.3 Random numbers (`np.rand`)

6.4 optimization (`np.optimization`)

7 Loading and handling data

Download notebook.

7.1 Loading data from files (`csv`, `xlsx`, `json`)

7.2 Loading data from online resources (`requests`)

7.3 Dataframes (`pandas`)

dt.strftime("%-m/%-d/%y, %H:%M")

7.4 json

7.5 csv

7.6 SQL databases (`sqlite3`)

8 Analyzing data

Download notebook.

8.1 Visualization (`matplotlib`)

8.2 csv, json, requests

8.3 dataframe

8.4 dealing with na

9 An example project

Download notebook.

Human population data

10 Miscellaneous

We collect here some stuff, which does not really fit anywhere else. We use the material in this section without notice in the whole course.

10.1 Comments

In Python, the key for a comment is #.

```
# This line is a comment.
print("This line is not a comment.")
```

```
This line is not a comment.
```

10.2 Two equality signs

In Python (and many other programming languages), there are assignments such as `x = 0`, which says that `x` should from now on have the value `0`, and `x == 0`. The latter is a condition, which results in `True` if `x` is indeed `0`, and 'False otherwise.

10.3 User input

Sometimes, when operating on the command line, one wants to ask for user input. This works as follows:

```
name = input("What is your name?")
print(f"Hello, {name}!")
```

10.4 Regular expressions

ask the AI!

10.5 dot-notation

Here is another very useful notation convention, known as dot-notation: For a function `str.something(s,...)`, where `s` is of type `str`, you can as well write `s.something()`. For example, `s.strip()` is the same as `str.strip(s)`.

10.6 git and github

These course notes are organized using the version control system `git`. If this does not mean anything to you, don't mind. At the moment, this only means that you need to install `git` on your system, and run in the folder where you want to have the course notes

```
git clone https://github.com/pfaffelh/python_for_data/
```

With this command, you are *cloning* the repository of the course notes from github. You can also look at this repository online.

There is a good chance that the course notes will be updated during the upcoming weeks. In order to obtain the most current version, you need to run

```
git pull
```

There is (at least) one catch for using `git`: If you are changing one of the course files locally on your system, but the file has updated, `git pull` realizes the different versions and rejects to do `git pull`. The best way out is to do all changes to the files in the folder `myFiles`, or only change the files `...ipynb`. All these files are excluded from any update with `git pull`.

We recommend that you do the following:

1. Clone the repository of these notes using

```
git clone https://github.com/pfaffelh/python_for_data
```

Hit `cd python_for_data`. 2. Create a virtual environment using `python -m venv venv` or `python3 -m venv venv`. 3. Activate this environment by `source venv/bin/activate`, 4. Install the libraries using `pip3 install -r requirements.txt` (or `pip install -r requirements.txt`).

11 NumPy for Machine Learning

A Lecture-Complete Course Script for Mathematicians

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

12 How to Use This Script

This document is designed as a lecture-complete course script for a mathematically trained audience. It aims to bridge rigorous mathematical thinking and practical numerical implementation.

The guiding principle of the course is:

> **Mathematics → vectorized formulation → NumPy implementation → numerical interpretation**

The intended audience is a group of students who are comfortable with linear algebra, calculus, basic proof-oriented reasoning, and elementary Python.

The central didactic choices are the following.

1. We treat **NumPy** as the computational language of finite-dimensional linear algebra.
2. We treat **pandas** as a data interface rather than as the mathematical core.
3. We implement fundamental machine learning models directly in NumPy before mentioning higher-level libraries.
4. We repeatedly connect formulas to shapes, dimensions, and numerical stability.

13 Learning Objectives

After working through this script, students should be able to:

- use NumPy arrays as vectors, matrices, and batches of data;
- reason confidently about shapes, axes, and broadcasting;
- distinguish elementwise operations from linear-algebraic operations;
- derive and implement linear regression from first principles;
- implement gradient descent and study its convergence behavior;
- derive and implement logistic regression for binary classification;
- understand the role of feature scaling and data preprocessing;
- implement a simple neural network using matrix operations;
- recognize common sources of numerical instability and shape errors;
- connect abstract mathematics with computational workflows.

14 Course Structure

The course is organized as follows.

1. NumPy as a language for vectors and matrices.
2. Shapes, slicing, views, copies, and the logic of vectorization.
3. Linear algebra in NumPy.
4. Statistics and aggregation.
5. pandas as a gateway from raw tabular data to numerical arrays.
6. Linear regression and the normal equation.
7. Gradient descent and optimization.
8. Logistic regression and binary cross-entropy.
9. Data preprocessing and feature engineering.
10. A first neural network in pure NumPy.
11. Numerical stability and debugging strategies.
12. Mini-projects and extensions.

15 Mathematical Convention

Throughout the text:

- vectors are written in lowercase, for example $x \in \mathbb{R}^d$;
- matrices are written in uppercase, for example $X \in \mathbb{R}^{n \times d}$;
- rows of X usually represent samples;
- columns of X usually represent features.

Thus a dataset with n observations and d features is represented by a matrix

$$X \in \mathbb{R}^{n \times d}.$$

A parameter vector for a linear model is usually written as

$$w \in \mathbb{R}^d.$$

Predictions then take the form

$$\hat{y} = Xw.$$

16 1. NumPy as a Computational Model of Linear Algebra

16.1 1.1 Arrays, vectors, matrices

NumPy arrays are the fundamental data structure of the course. A one-dimensional array behaves like a vector, and a two-dimensional array behaves like a matrix.

```
x = np.array([1.0, 2.0, 3.0])
A = np.array([
    [1.0, 0.0, 2.0],
    [0.0, 1.0, 1.0]
])

print("x =")
print(x)
print()
print("A =")
print(A)
print()
print("x.shape =", x.shape)
print("A.shape =", A.shape)
print("x.ndim =", x.ndim)
print("A.ndim =", A.ndim)
print("x.dtype =", x.dtype)
```

```
x =
[1. 2. 3.]

A =
[[1. 0. 2.]
 [0. 1. 1.]]

x.shape = (3,)
A.shape = (2, 3)
```

```
x.ndim = 1
A.ndim = 2
x.dtype = float64
```

A first important observation is that the shape of a vector in NumPy is not `(3, 1)` but `(3,)`. This is mathematically convenient in some places and dangerous in others. Much of good NumPy programming consists in keeping track of such distinctions.

16.2 1.2 Column vectors versus one-dimensional arrays

From a mathematical viewpoint, one may want to distinguish between a vector in $\mathbb{R}^d$ and a column vector in $\mathbb{R}^{d\times 1}$. NumPy does not force that distinction unless we create it explicitly.

```
x = np.array([1.0, 2.0, 3.0])
x_col = x.reshape(-1, 1)
x_row = x.reshape(1, -1)

print("x.shape =", x.shape)
print("x_col.shape =", x_col.shape)
print("x_row.shape =", x_row.shape)
print()
print("x_col =")
print(x_col)
print()
print("x_row =")
print(x_row)
```

```
x.shape = (3,)
x_col.shape = (3, 1)
x_row.shape = (1, 3)

x_col =
[[1.]
 [2.]
 [3.]]

x_row =
[[1. 2. 3.]]
```

This is one of the most important habits for students: always ask what the shape of each object is.

16.3 1.3 Basic arithmetic

```
x = np.array([1.0, -2.0, 4.0])
y = np.array([3.0, 5.0, -1.0])

print("x + y =", x + y)
print("x - y =", x - y)
print("2 * x =", 2 * x)
print("x / 2 =", x / 2)
print("x ** 2 =", x ** 2)
```

```
x + y = [4. 3. 3.]
x - y = [-2. -7.  5.]
2 * x = [ 2. -4.  8.]
x / 2 = [ 0.5 -1.   2. ]
x ** 2 = [ 1.  4. 16.]
```

These are **elementwise** operations. That distinction matters. Elementwise multiplication is not the same as matrix multiplication.

16.4 1.4 Inner product, norms, and basic geometry

```
x = np.array([1.0, -2.0, 2.0])
y = np.array([3.0, 1.0, -1.0])

inner_1 = x @ y
inner_2 = np.dot(x, y)
norm_l1 = np.sum(np.abs(x))
norm_l2 = np.sqrt(np.sum(x**2))
norm_l2_alt = np.linalg.norm(x)
norm_inf = np.max(np.abs(x))

print("inner_1 =", inner_1)
print("inner_2 =", inner_2)
print("||x||_1 =", norm_l1)
print("||x||_2 =", norm_l2)
print("||x||_2 (np.linalg.norm) =", norm_l2_alt)
print("||x||_inf =", norm_inf)
```

```
inner_1 = -1.0
inner_2 = -1.0
||x||_1 = 5.0
||x||_2 = 3.0
||x||_2 (np.linalg.norm) = 3.0
||x||_inf = 2.0
```

The formula

$$\langle x, y \rangle = \sum_{j=1}^{d} x_j y_j$$

becomes either `x @ y` or `np.dot(x, y)`.

The Euclidean norm

$$\|x\|_2 = \sqrt{\langle x, x \rangle}$$

can be implemented by hand or by a library function. In a course for mathematicians it is worth showing both.

16.5 1.5 Exercise set

1. Create a vector $x \in \mathbb{R}^5$ of your choice.
2. Compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$.
3. Create another vector $y \in \mathbb{R}^5$ and compute the inner product.
4. Reshape x into a column vector and a row vector.
5. Explain in words why `x.shape == (5,)` is not the same as `x.reshape(5, 1).shape == (5, 1)`.

16.6 1.6 Worked solution sketch

```python
x = np.array([1.0, -1.0, 2.0, 0.0, 3.0])
y = np.array([2.0, 1.0, -1.0, 4.0, 0.5])

print("||x||_1 =", np.sum(np.abs(x)))
print("||x||_2 =", np.linalg.norm(x))
print("||x||_inf =", np.max(np.abs(x)))
```

```python
print("<x,y> =", x @ y)
print("column shape =", x.reshape(-1, 1).shape)
print("row shape =", x.reshape(1, -1).shape)
```

```
||x||_1 = 7.0
||x||_2 = 3.872983346207417
||x||_inf = 3.0
<x,y> = 0.5
column shape = (5, 1)
row shape = (1, 5)
```

17 2. Indexing, Slicing, Views, and Copies

17.1 2.1 One-dimensional indexing

```
a = np.arange(10)
print("a =", a)
print("a[0] =", a[0])
print("a[3] =", a[3])
print("a[-1] =", a[-1])
print("a[2:7] =", a[2:7])
print("a[:4] =", a[:4])
print("a[::2] =", a[::2])
```

```
a = [0 1 2 3 4 5 6 7 8 9]
a[0] = 0
a[3] = 3
a[-1] = 9
a[2:7] = [2 3 4 5 6]
a[:4] = [0 1 2 3]
a[::2] = [0 2 4 6 8]
```

17.2 2.2 Two-dimensional indexing

```
A = np.arange(1, 13).reshape(3, 4)
print("A =")
print(A)
print()
print("A[0, 0] =", A[0, 0])
print("A[1, 2] =", A[1, 2])
print("A[:, 1] =", A[:, 1])
print("A[1, :] =", A[1, :])
print("A[:2, 1:3] =")
print(A[:2, 1:3])
```

```
A =
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]

A[0, 0] = 1
A[1, 2] = 7
A[:, 1] = [ 2  6 10]
A[1, :] = [5 6 7 8]
A[:2, 1:3] =
[[2 3]
 [6 7]]
```

For a matrix $A \in \mathbb{R}^{m \times n}$, the expression `A[:, j]` selects the j-th column, whereas `A[i, :]` selects the i-th row.

17.3 2.3 Views versus copies

This topic is essential. A slice often creates a view rather than an independent object.

```python
a = np.arange(10)
b = a[2:6]

print("Before modification")
print("a =", a)
print("b =", b)

b[0] = 99

print()
print("After modification")
print("a =", a)
print("b =", b)
```

```
Before modification
a = [0 1 2 3 4 5 6 7 8 9]
b = [2 3 4 5]

After modification
a = [ 0  1 99  3  4  5  6  7  8  9]
b = [99  3  4  5]
```

The modification of `b` also changed `a`. This means that `b` was a view on part of `a`.

To force an actual copy, use `.copy()`.

```
a = np.arange(10)
c = a[2:6].copy()
c[0] = -7

print("a =", a)
print("c =", c)
```

```
a = [0 1 2 3 4 5 6 7 8 9]
c = [-7  3  4  5]
```

17.4 2.4 Boolean masks

Boolean indexing is a central tool for data preprocessing.

```
x = np.array([3.0, -1.0, 2.5, -4.0, 0.0, 7.0])
mask = x > 0

print("x =", x)
print("mask =", mask)
print("x[mask] =", x[mask])
```

```
x = [ 3.  -1.   2.5 -4.   0.   7. ]
mask = [ True False  True False False  True]
x[mask] = [3.  2.5 7. ]
```

For matrices the same logic works rowwise or elementwise depending on the mask.

```
X = np.array([
    [1.0, 10.0],
    [-1.0, 20.0],
    [2.0, 30.0],
    [-2.0, 40.0]
])

row_mask = X[:, 0] > 0
print("row_mask =", row_mask)
```

```
print("Filtered rows =")
print(X[row_mask])
```

```
row_mask = [ True False  True False]
Filtered rows =
[[ 1. 10.]
 [ 2. 30.]]
```

17.5 2.5 Exercises

1. Create a 4×5 matrix and select its second row and third column.
2. Create a slice of the matrix and verify whether modifying the slice changes the original matrix.
3. Create a copy of the slice and verify that modifying it does not change the original matrix.
4. Use boolean indexing to select all entries of a vector that are strictly larger than 1.
5. Filter all rows of a matrix whose first component is positive.

17.6 2.6 Worked example

```
A = np.arange(20).reshape(4, 5)
print("A =")
print(A)
print()
print("second row =", A[1, :])
print("third column =", A[:, 2])

sub = A[:2, :3]
sub[0, 0] = -100
print()
print("A after modifying a view =")
print(A)

A = np.arange(20).reshape(4, 5)
sub_copy = A[:2, :3].copy()
sub_copy[0, 0] = -100
print()
print("A after modifying a copy =")
print(A)
```

```python
print("sub_copy =")
print(sub_copy)
```

```
A =
[[ 0  1  2  3  4]
 [ 5  6  7  8  9]
 [10 11 12 13 14]
 [15 16 17 18 19]]

second row = [5 6 7 8 9]
third column = [ 2  7 12 17]

A after modifying a view =
[[-100    1    2    3    4]
 [   5    6    7    8    9]
 [  10   11   12   13   14]
 [  15   16   17   18   19]]

A after modifying a copy =
[[ 0  1  2  3  4]
 [ 5  6  7  8  9]
 [10 11 12 13 14]
 [15 16 17 18 19]]
sub_copy =
[[-100    1    2]
 [   5    6    7]]
```

18 3. Vectorization and Broadcasting

18.1 3.1 Why vectorization matters

In machine learning we rarely want to process one data point at a time in pure Python loops. Instead we formulate operations on full vectors and matrices.

Suppose we want to evaluate the affine function

$$f(x) = 3x + 2$$

at many points simultaneously.

```
x = np.linspace(0.0, 1.0, 6)
y = 3 * x + 2

print("x =", x)
print("y =", y)
```

```
x = [0.  0.2 0.4 0.6 0.8 1. ]
y = [2.  2.6 3.2 3.8 4.4 5. ]
```

This is both shorter and faster than a Python loop.

18.2 3.2 Loop versus vectorized formulation

```
x = np.linspace(0.0, 1.0, 6)

y_loop = np.empty_like(x)
for i in range(len(x)):
    y_loop[i] = 3 * x[i] + 2

y_vec = 3 * x + 2
```

```
print("y_loop =", y_loop)
print("y_vec  =", y_vec)
print("difference =", np.max(np.abs(y_loop - y_vec)))
```

```
y_loop = [2.  2.6 3.2 3.8 4.4 5. ]
y_vec  = [2.  2.6 3.2 3.8 4.4 5. ]
difference = 0.0
```

The vectorized formulation exposes the mathematics more clearly: the expression looks like the formula itself.

18.3 3.3 Broadcasting idea

Broadcasting allows NumPy to combine arrays of compatible shape even when their shapes are not identical.

```
X = np.array([
    [1.0, 2.0],
    [3.0, 4.0],
    [5.0, 6.0]
])

b = np.array([10.0, 20.0])

print("X =")
print(X)
print()
print("b =", b)
print()
print("X + b =")
print(X + b)
```

```
X =
[[1. 2.]
 [3. 4.]
 [5. 6.]]

b = [10. 20.]
```

```
X + b =
[[11. 22.]
 [13. 24.]
 [15. 26.]]
```

The vector `b` is broadcast across all rows.

From the machine learning viewpoint this is exactly what happens when one adds a bias vector to a batch of hidden activations or feature values.

18.4 3.4 Broadcasting with column vectors

```
x = np.array([1.0, 2.0, 3.0])
x_col = x[:, None]
x_row = x[None, :]

print("x_col =")
print(x_col)
print()
print("x_row =")
print(x_row)
print()
print("x_col + x_row =")
print(x_col + x_row)
```

```
x_col =
[[1.]
 [2.]
 [3.]]

x_row =
[[1. 2. 3.]]

x_col + x_row =
[[2. 3. 4.]
 [3. 4. 5.]
 [4. 5. 6.]]
```

This creates the matrix whose (i, j)-entry is $x_i + x_j$. In other words, broadcasting can be used to construct pairwise operations compactly.

18.5 3.5 A distance-matrix example

For a set of one-dimensional points $x_1, \dots, x_n$, the matrix

$$D_{ij} = |x_i - x_j|$$

can be built without loops.

```
x = np.array([0.0, 1.5, 3.0, 5.0])
D = np.abs(x[:, None] - x[None, :])

print("x =", x)
print("distance matrix =")
print(D)
```

```
x = [0.  1.5 3.  5. ]
distance matrix =
[[0.  1.5 3.  5. ]
 [1.5 0.  1.5 3.5]
 [3.  1.5 0.  2. ]
 [5.  3.5 2.  0. ]]
```

18.6 3.6 Batch affine maps

If $X \in \mathbb{R}^{n \times d}$ is a dataset and $w \in \mathbb{R}^d$, then

$$Xw$$

computes the predictions for all n points at once.

```
X = np.array([
    [1.0, 2.0],
    [0.0, 1.0],
    [3.0, 1.0]
])
w = np.array([2.0, -1.0])
b = 0.5

y = X @ w + b
print("y =", y)
```

```
y = [ 0.5 -0.5  5.5]
```

This is one of the core patterns of the entire course.

18.7 3.7 Exercises

1. Re-implement a simple vectorized formula using a Python loop.
2. Add a vector `b` to each row of a matrix `X`.
3. Construct a matrix of pairwise differences using broadcasting.
4. Given a data matrix X and parameter vector w, compute $Xw + b$.
5. Explain in words what broadcasting is doing in each of the examples above.

18.8 3.8 Worked example

```python
X = np.arange(12, dtype=float).reshape(4, 3)
b = np.array([10.0, 20.0, 30.0])

print("X =")
print(X)
print()
print("X + b =")
print(X + b)

u = np.array([1.0, 4.0, 6.0, 8.0])
pairwise = u[:, None] - u[None, :]
print()
print("pairwise differences =")
print(pairwise)
```

```
X =
[[ 0.  1.  2.]
 [ 3.  4.  5.]
 [ 6.  7.  8.]
 [ 9. 10. 11.]]

X + b =
[[10. 21. 32.]
 [13. 24. 35.]
```

```
 [16. 27. 38.]
 [19. 30. 41.]]

pairwise differences =
[[ 0. -3. -5. -7.]
 [ 3.  0. -2. -4.]
 [ 5.  2.  0. -2.]
 [ 7.  4.  2.  0.]]
```

19 4. Matrix Calculus in NumPy

19.1 4.1 Matrix-vector products

```
A = np.array([
    [1.0, 2.0],
    [3.0, 4.0]
])
x = np.array([1.0, -1.0])

print("A @ x =", A @ x)
```

```
A @ x = [-1. -1.]
```

The formula

$$(Ax)_i = \sum_{j=1}^{n} A_{ij} x_j$$

becomes the very compact expression `A @ x`.

19.2 4.2 Matrix-matrix products

```
A = np.array([
    [1.0, 2.0],
    [3.0, 4.0]
])
B = np.array([
    [2.0, 0.0],
    [1.0, 2.0]
])
```

```
print("A @ B =")
print(A @ B)
```

```
A @ B =
[[ 4.  4.]
 [10.  8.]]
```

19.3 4.3 Transposition

```
A = np.array([
    [1.0, 2.0, 3.0],
    [4.0, 5.0, 6.0]
])

print("A =")
print(A)
print()
print("A.T =")
print(A.T)
```

```
A =
[[1. 2. 3.]
 [4. 5. 6.]]

A.T =
[[1. 4.]
 [2. 5.]
 [3. 6.]]
```

The transpose changes rows into columns. In optimization and machine learning, transposes appear everywhere: normal equations, gradients, covariance matrices, and backpropagation.

19.4 4.4 Solving linear systems

If we want to solve

$$Ax = b,$$

then `np.linalg.solve` is the standard tool.

```
A = np.array([
    [3.0, 1.0],
    [1.0, 2.0]
])
b = np.array([9.0, 8.0])

x = np.linalg.solve(A, b)
print("solution x =", x)
print("A @ x =", A @ x)
```

```
solution x = [2. 3.]
A @ x = [9. 8.]
```

19.5 4.5 Why one should usually avoid explicit inverses

One often sees formulas like

$$x = A^{-1}b.$$

This is mathematically correct when A is invertible, but numerically it is often better to solve the system directly.

```
A = np.array([
    [3.0, 1.0],
    [1.0, 2.0]
])
b = np.array([9.0, 8.0])

x_solve = np.linalg.solve(A, b)
x_inv = np.linalg.inv(A) @ b

print("x_solve =", x_solve)
print("x_inv   =", x_inv)
print("difference =", np.linalg.norm(x_solve - x_inv))
```

```
x_solve = [2. 3.]
x_inv   = [2. 3.]
difference = 2.220446049250313e-16
```

The differences are small here, but in more difficult problems direct solving is preferred.

19.6 4.6 Condition number

The condition number indicates how sensitive a linear system is to perturbations.

```
A = np.array([
    [1.0, 1.0],
    [1.0, 1.0001]
])

cond = np.linalg.cond(A)
print("condition number =", cond)
```

```
condition number = 40002.00007491187
```

A large condition number warns us that the problem may be numerically unstable.

19.7 4.7 Eigenvalues and singular values

Although not needed immediately, it is useful to know that NumPy also provides tools for spectral problems.

```
A = np.array([
    [2.0, 1.0],
    [1.0, 2.0]
])

eigenvalues, eigenvectors = np.linalg.eig(A)
U, S, VT = np.linalg.svd(A)

print("eigenvalues =", eigenvalues)
print("singular values =", S)
```

```
eigenvalues = [3. 1.]
singular values = [3. 1.]
```

19.8 4.8 Exercises

1. Compute a matrix-vector product by hand and verify it in NumPy.
2. Solve a 2×2 linear system using `np.linalg.solve`.
3. Compare `np.linalg.solve(A, b)` with `np.linalg.inv(A) @ b`.
4. Construct an ill-conditioned matrix and compute its condition number.
5. Compute the eigenvalues of a symmetric matrix and interpret them geometrically.

19.9 4.9 Worked example

```
A = np.array([
    [4.0, 1.0],
    [1.0, 3.0]
])
b = np.array([1.0, 2.0])

x = np.linalg.solve(A, b)
print("x =", x)
print("residual =", np.linalg.norm(A @ x - b))

eigvals, eigvecs = np.linalg.eig(A)
print("eigvals =", eigvals)
print("eigvecs =")
print(eigvecs)
```

```
x = [0.09090909 0.63636364]
residual = 0.0
eigvals = [4.61803399 2.38196601]
eigvecs =
[[ 0.85065081 -0.52573111]
 [ 0.52573111  0.85065081]]
```

20 5. Aggregation, Statistics, and the `axis` Parameter

20.1 5.1 Means and sums

Statistics is everywhere in machine learning: means, variances, losses, empirical risks, and error metrics.

```
data = np.array([
    [1.0, 2.0, 3.0],
    [2.0, 3.0, 4.0],
    [3.0, 4.0, 5.0]
])

print("sum over all entries =", np.sum(data))
print("mean over all entries =", np.mean(data))
```

```
sum over all entries = 27.0
mean over all entries = 3.0
```

20.2 5.2 Means along an axis

```
print("column means =", np.mean(data, axis=0))
print("row means =", np.mean(data, axis=1))
```

```
column means = [2. 3. 4.]
row means = [2. 3. 4.]
```

The meaning of `axis` is worth internalizing.

- `axis=0` means aggregate down the rows, so the result keeps one entry per column.
- `axis=1` means aggregate across the columns, so the result keeps one entry per row.

20.3 5.3 Variance and standard deviation

```
print("column variances =", np.var(data, axis=0))
print("column std =", np.std(data, axis=0))
```

```
column variances = [0.66666667 0.66666667 0.66666667]
column std = [0.81649658 0.81649658 0.81649658]
```

In practical machine learning, these are central for normalization and data exploration.

20.4 5.4 Standardization

Given a feature matrix X, one often transforms each column by

$$\widetilde{X}_{:,j} = \frac{X_{:,j} - \mu_j}{\sigma_j}.$$

```
X = np.array([
    [1.0, 10.0],
    [2.0, 20.0],
    [3.0, 30.0],
    [4.0, 40.0]
])

mu = np.mean(X, axis=0)
sigma = np.std(X, axis=0)
X_std = (X - mu) / sigma

print("mu =", mu)
print("sigma =", sigma)
print("X_std =")
print(X_std)
print("mean of standardized columns =", np.mean(X_std, axis=0))
print("std of standardized columns =", np.std(X_std, axis=0))
```

```
mu = [ 2.5 25. ]
sigma = [ 1.11803399 11.18033989]
X_std =
```

```
[[-1.34164079 -1.34164079]
 [-0.4472136  -0.4472136 ]
 [ 0.4472136   0.4472136 ]
 [ 1.34164079  1.34164079]]
mean of standardized columns = [0. 0.]
std of standardized columns = [1. 1.]
```

20.5 5.5 Loss functions as aggregations

For regression, the mean squared error is

$$\mathrm{MSE}(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

```python
y = np.array([2.0, 3.0, 5.0, 4.0])
y_pred = np.array([2.2, 2.8, 4.7, 4.3])

mse = np.mean((y - y_pred)**2)
mae = np.mean(np.abs(y - y_pred))

print("MSE =", mse)
print("MAE =", mae)
```

```
MSE = 0.06499999999999999
MAE = 0.25
```

20.6 5.6 Exercises

1. Compute the row sums and column sums of a matrix.
2. Standardize a small matrix and verify the result numerically.
3. Implement the mean squared error by hand.
4. Compare MSE and MAE on the same prediction vector.
5. Explain why standardization is done featurewise rather than over all entries at once.

20.7 5.7 Worked example

```python
X = np.array([
    [2.0, 1.0, 3.0],
    [4.0, 2.0, 6.0],
    [6.0, 3.0, 9.0]
])

print("column sums =", np.sum(X, axis=0))
print("row sums =", np.sum(X, axis=1))

mu = np.mean(X, axis=0)
sigma = np.std(X, axis=0)
Z = (X - mu) / sigma

print("Z =")
print(Z)
print("column means of Z =", np.mean(Z, axis=0))
print("column std of Z =", np.std(Z, axis=0))
```

```
column sums = [12.  6. 18.]
row sums = [ 6. 12. 18.]
Z =
[[-1.22474487 -1.22474487 -1.22474487]
 [ 0.          0.          0.        ]
 [ 1.22474487  1.22474487  1.22474487]]
column means of Z = [0. 0. 0.]
column std of Z = [1. 1. 1.]
```

21 6. Randomness, Reproducibility, and Synthetic Data

21.1 6.1 Why randomness matters

Randomness enters machine learning in many places:

- synthetic data generation;
- random initialization of parameters;
- train-test splitting;
- mini-batch sampling;
- randomized algorithms.

21.2 6.2 Modern random number generation

The recommended interface is `np.random.default_rng`.

```
rng = np.random.default_rng(42)

print("uniform random numbers =", rng.random(5))
print("normal random numbers =", rng.standard_normal(5))
print("random integers =", rng.integers(0, 10, size=5))
```

```
uniform random numbers = [0.77395605 0.43887844 0.85859792 0.69736803 0.09417735]
normal random numbers = [-1.30217951  0.1278404  -0.31624259 -0.01680116 -
0.85304393]
random integers = [5 3 1 9 7]
```

21.3 6.3 Reproducibility

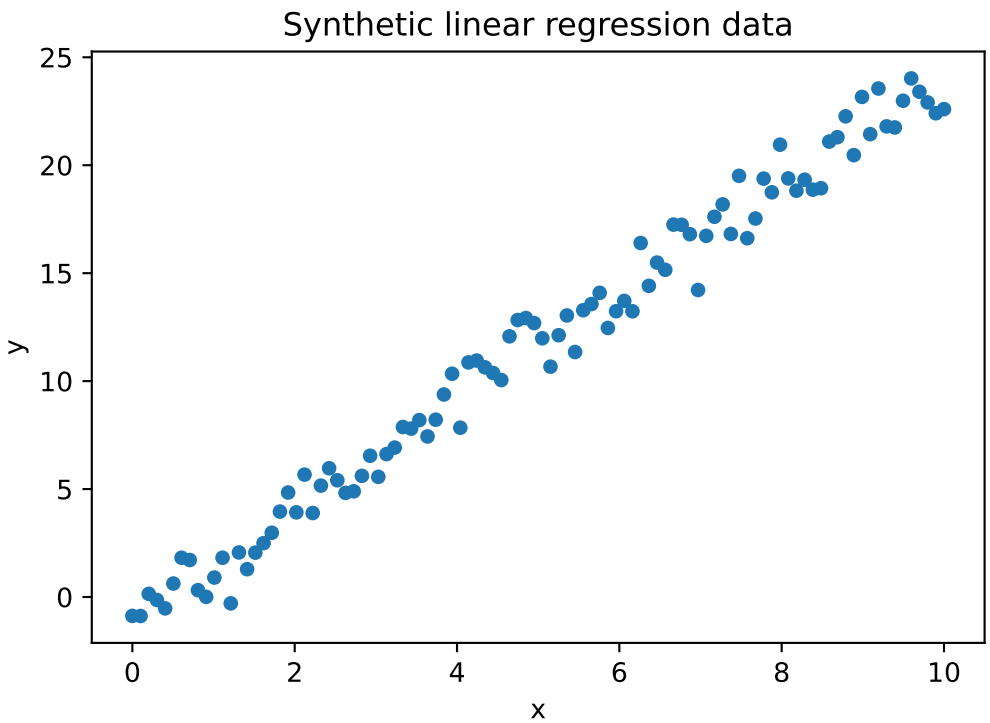
```
rng1 = np.random.default_rng(123)
rng2 = np.random.default_rng(123)

print(rng1.standard_normal(4))
print(rng2.standard_normal(4))
```

```
[-0.98912135 -0.36778665  1.28792526  0.19397442]
[-0.98912135 -0.36778665  1.28792526  0.19397442]
```

The outputs agree because the seeds agree.

21.4 6.4 A synthetic linear regression dataset

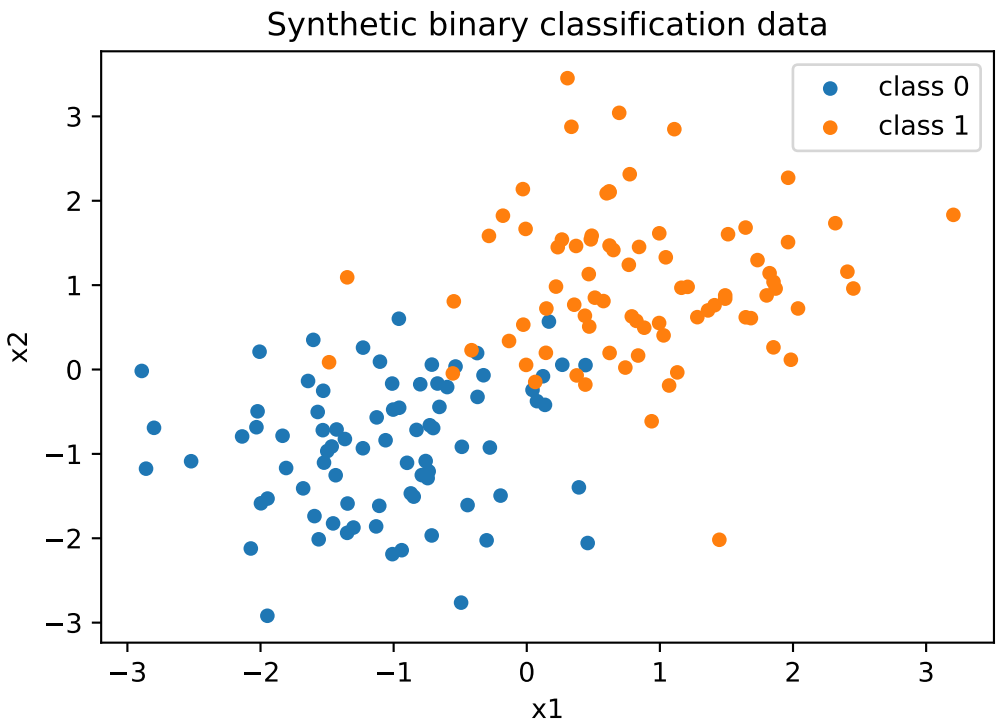
```
rng = np.random.default_rng(0)

n = 100
x = np.linspace(0.0, 10.0, n)
noise = rng.normal(0.0, 1.0, size=n)
y = 2.5 * x - 1.0 + noise

plt.figure(figsize=(6, 4))
plt.scatter(x, y, s=20)
plt.xlabel("x")
plt.ylabel("y")
plt.title("Synthetic linear regression data")
plt.show()
```



21.5 6.5 A synthetic classification dataset

```
rng = np.random.default_rng(0)

n0 = 80
n1 = 80

X0 = rng.normal(loc=[-1.0, -1.0], scale=0.8, size=(n0, 2))
X1 = rng.normal(loc=[1.0, 1.0], scale=0.8, size=(n1, 2))

X = np.vstack([X0, X1])
y = np.concatenate([np.zeros(n0), np.ones(n1)])

plt.figure(figsize=(6, 4))
plt.scatter(X0[:, 0], X0[:, 1], label="class 0", s=20)
plt.scatter(X1[:, 0], X1[:, 1], label="class 1", s=20)
plt.legend()
plt.xlabel("x1")
```

```
plt.ylabel("x2")
plt.title("Synthetic binary classification data")
plt.show()
```



21.6 6.6 Train-test splitting by permutation

```
rng = np.random.default_rng(7)
n = len(X)
perm = rng.permutation(n)

train_size = int(0.8 * n)
train_idx = perm[:train_size]
test_idx = perm[train_size:]

X_train = X[train_idx]
X_test = X[test_idx]
y_train = y[train_idx]
```

```
y_test = y[test_idx]

print("X_train.shape =", X_train.shape)
print("X_test.shape =", X_test.shape)
```

```
X_train.shape = (128, 2)
X_test.shape = (32, 2)
```

21.7 6.7 Exercises

1. Generate a random vector with a fixed seed and verify reproducibility.
2. Create a synthetic regression dataset with noise.
3. Create a synthetic binary classification dataset in two dimensions.
4. Split a dataset into train and test subsets using a permutation.
5. Explain why fixed seeds are useful in teaching and experimentation.

21.8 6.8 Worked example

```
rng = np.random.default_rng(2024)

X = rng.normal(size=(12, 3))
y = (X[:, 0] - X[:, 1] > 0).astype(int)

perm = rng.permutation(len(X))
split = 8

X_train = X[perm[:split]]
X_test = X[perm[split:]]
y_train = y[perm[:split]]
y_test = y[perm[split:]]

print("y_train =", y_train)
print("y_test =", y_test)
```

```
y_train = [1 0 1 1 1 0 0 0]
y_test = [0 1 0 1]
```

22 7. pandas as a Data Interface

22.1 7.1 Why pandas belongs after NumPy

For mathematically trained students it is usually more effective to learn NumPy first. Once arrays, shapes, and matrix operations are conceptually clear, pandas can be introduced as an interface for real-world tabular data.

The key pedagogical message is:

> **pandas handles labeled tabular data; NumPy performs the mathematical core.**

22.2 7.2 A small example DataFrame

```
df = pd.DataFrame({
    "age": [22, 27, 31, 45, 38, 29],
    "income": [42000, 51000, 64000, 83000, 72000, 56000],
    "hours_studied": [10, 12, 7, 4, 6, 9],
    "passed": [1, 1, 0, 0, 1, 1]
})

df
```

	age	income	hours_studied	passed
0	22	42000	10	1
1	27	51000	12	1
2	31	64000	7	0
3	45	83000	4	0
4	38	72000	6	1
5	29	56000	9	1

A DataFrame can be viewed as a matrix together with labels and heterogeneous metadata.

22.3 7.3 Selecting columns and rows

```
print(df["age"])
print()
print(df[["age", "income"]])
print()
print(df.iloc[0])
print()
print(df.loc[0:2, ["age", "passed"]])
```

```
0    22
1    27
2    31
3    45
4    38
5    29
Name: age, dtype: int64

   age  income
0   22   42000
1   27   51000
2   31   64000
3   45   83000
4   38   72000
5   29   56000

age                 22
income           42000
hours_studied       10
passed               1
Name: 0, dtype: int64

   age  passed
0   22       1
1   27       1
2   31       0
```

22.4 7.4 Filtering rows

```
filtered = df[df["income"] > 55000]
filtered
```

	age	income	hours_studied	passed
2	31	64000	7	0
3	45	83000	4	0
4	38	72000	6	1
5	29	56000	9	1

22.5 7.5 Missing values

```
df_missing = df.copy()
df_missing.loc[2, "income"] = np.nan
df_missing.loc[4, "hours_studied"] = np.nan

print(df_missing)
print()
print(df_missing.isna())
print()
print(df_missing.dropna())
```

```
   age   income  hours_studied  passed
0   22  42000.0           10.0       1
1   27  51000.0           12.0       1
2   31      NaN            7.0       0
3   45  83000.0            4.0       0
4   38  72000.0            NaN       1
5   29  56000.0            9.0       1

     age  income  hours_studied  passed
0  False   False          False   False
1  False   False          False   False
2  False    True          False   False
3  False   False          False   False
4  False   False           True   False
```

```
5  False   False          False   False

   age   income  hours_studied  passed
0   22  42000.0           10.0       1
1   27  51000.0           12.0       1
3   45  83000.0            4.0       0
5   29  56000.0            9.0       1
```

22.6 7.6 From pandas to NumPy

This is the crucial transition.

```
clean = df_missing.dropna()

X = clean[["age", "income", "hours_studied"]].to_numpy(dtype=float)
y = clean["passed"].to_numpy(dtype=float)

print("X =")
print(X)
print()
print("y =", y)
print("X.shape =", X.shape)
print("y.shape =", y.shape)
```

```
X =
[[2.2e+01 4.2e+04 1.0e+01]
 [2.7e+01 5.1e+04 1.2e+01]
 [4.5e+01 8.3e+04 4.0e+00]
 [2.9e+01 5.6e+04 9.0e+00]]

y = [1. 1. 0. 1.]
X.shape = (4, 3)
y.shape = (4,)
```

Once we reach `X` and `y`, we are back in the mathematical world of arrays.

22.7 7.7 Reading a CSV file

The next cell assumes that a file named `ml_dataset.csv` is available in the same directory. If that file is not present, the fallback code creates a simple example dataset instead.

```python
from pathlib import Path

csv_path = Path("ml_dataset.csv")

if csv_path.exists():
    df_csv = pd.read_csv(csv_path)
else:
    rng = np.random.default_rng(42)
    n = 120
    df_csv = pd.DataFrame({
        "age": rng.integers(18, 65, size=n),
        "income": rng.normal(55000, 14000, size=n),
        "hours_studied": rng.normal(8, 2, size=n),
    })
    logits = (
        -4
        + 0.00005 * df_csv["income"]
        + 0.25 * df_csv["hours_studied"]
        - 0.02 * df_csv["age"]
    )
    probs = 1 / (1 + np.exp(-logits))
    df_csv["target"] = (rng.random(n) < probs).astype(int)

df_csv.head()
```

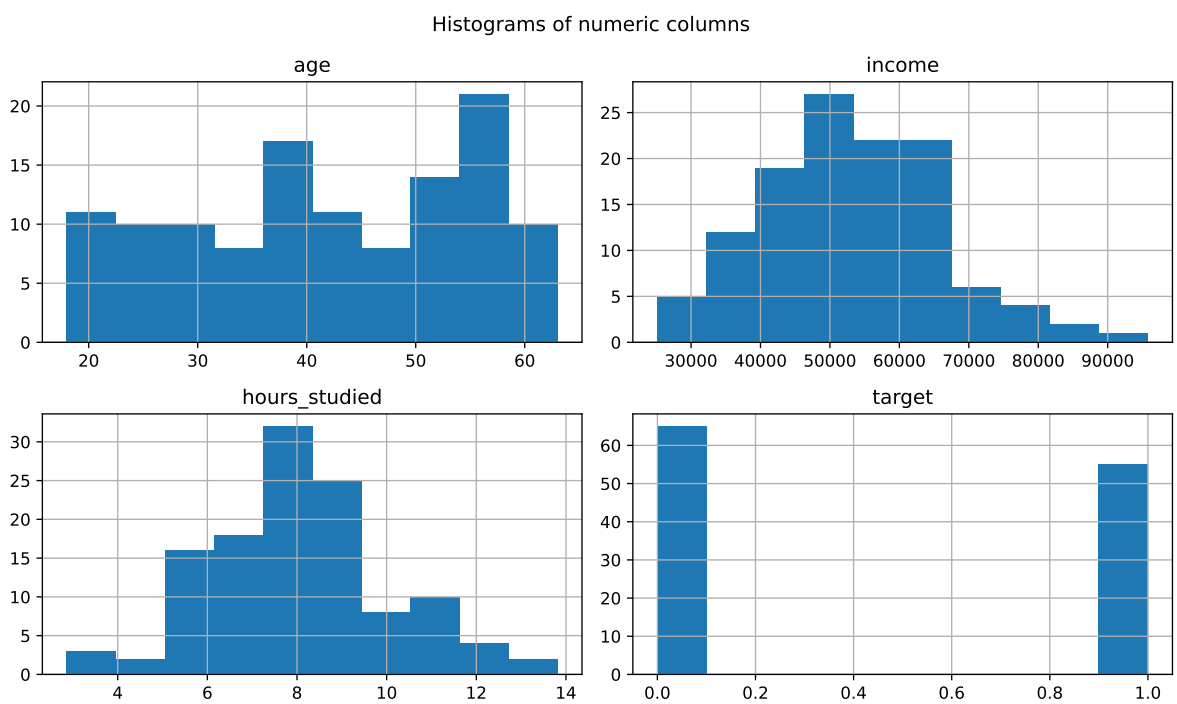
	age	income	hours_studied	target
0	22	42878.364380	7.762723	0
1	54	68555.896964	8.571652	1
2	48	31439.823197	10.612003	1
3	38	50311.609580	8.438765	1
4	38	57278.542911	7.178146	1

22.8 7.8 Basic exploratory analysis

```python
print(df_csv.describe())
print()
print(df_csv.dtypes)
```

```
             age        income  hours_studied      target
count  120.00000    120.000000     120.000000  120.000000
mean    41.95000  53343.380771       7.991499    0.458333
std     12.83499  12735.834218       2.016080    0.500350
min     18.00000  25151.352070       2.866683    0.000000
25%     31.00000  43345.030963       6.670809    0.000000
50%     42.00000  52550.425867       7.808451    0.000000
75%     54.00000  61287.290680       8.879359    1.000000
max     63.00000  95794.074524      13.810134    1.000000

age                int64
income           float64
hours_studied    float64
target             int64
dtype: object
```

```
numeric_cols = [c for c in df_csv.columns if pd.api.types.is_numeric_dtype(df_csv[c])]
df_csv[numeric_cols].hist(figsize=(10, 6))
plt.suptitle("Histograms of numeric columns")
plt.tight_layout()
plt.show()
```



22.9 7.9 Exercises

1. Create a DataFrame with at least three columns and six rows.
2. Select two columns and convert them to a NumPy array.
3. Introduce missing values and remove them.
4. Filter a DataFrame by a threshold condition.
5. Explain the sentence "pandas is the interface, NumPy is the engine."

22.10 7.10 Worked example

```
df_small = pd.DataFrame({
    "feature_1": [1.0, 2.0, 3.0, 4.0],
    "feature_2": [10.0, 20.0, 30.0, 40.0],
    "label": [0, 1, 0, 1]
})

X = df_small[["feature_1", "feature_2"]].to_numpy()
y = df_small["label"].to_numpy()

print("X =")
print(X)
print("y =", y)
```

```
X =
[[ 1. 10.]
 [ 2. 20.]
 [ 3. 30.]
 [ 4. 40.]]
y = [0 1 0 1]
```

23 8. Linear Regression from First Principles

23.1 8.1 Problem statement

Suppose we are given data

$$(x_i, y_i) \in \mathbb{R}^d \times \mathbb{R}, \quad i = 1, \dots, n.$$

We collect the inputs as rows of a matrix

$$X \in \mathbb{R}^{n \times d}$$

and the outputs as a vector

$$y \in \mathbb{R}^n.$$

A linear model predicts

$$\hat{y} = Xw.$$

The least squares objective is

$$f(w) = \|Xw - y\|_2^2.$$

Our goal is to find

$$w^* = \arg\min_{w \in \mathbb{R}^d} \|Xw - y\|_2^2.$$

23.2 8.2 Derivation of the normal equation

Expanding the square gives

$$f(w) = (Xw - y)^\top (Xw - y).$$

Differentiating with respect to w yields

$$\nabla f(w) = 2X^\top (Xw - y).$$

At a critical point we therefore require

$$X^\top (Xw - y) = 0,$$

hence

$$X^\top Xw = X^\top y.$$

This is the **normal equation**.

If $X^\top X$ is invertible, the minimizer is

$$w^* = (X^\top X)^{-1} X^\top y.$$

Computationally, we prefer the solve form

$$w^* = \mathrm{solve}(X^\top X,\ X^\top y).$$

23.3 8.3 A small one-dimensional example

```
X_raw = np.array([1.0, 2.0, 3.0, 4.0, 5.0])
y = np.array([1.8, 2.9, 3.7, 5.1, 5.9])

X = np.column_stack([np.ones(len(X_raw)), X_raw])

print("X =")
print(X)
print("y =", y)
```

```
X =
[[1. 1.]
 [1. 2.]
 [1. 3.]
 [1. 4.]
 [1. 5.]]
y = [1.8 2.9 3.7 5.1 5.9]
```

The first column of ones corresponds to the intercept term.

23.4 8.4 Solving the normal equation

```python
w = np.linalg.solve(X.T @ X, X.T @ y)
print("w =", w)
```

```
w = [0.76 1.04]
```

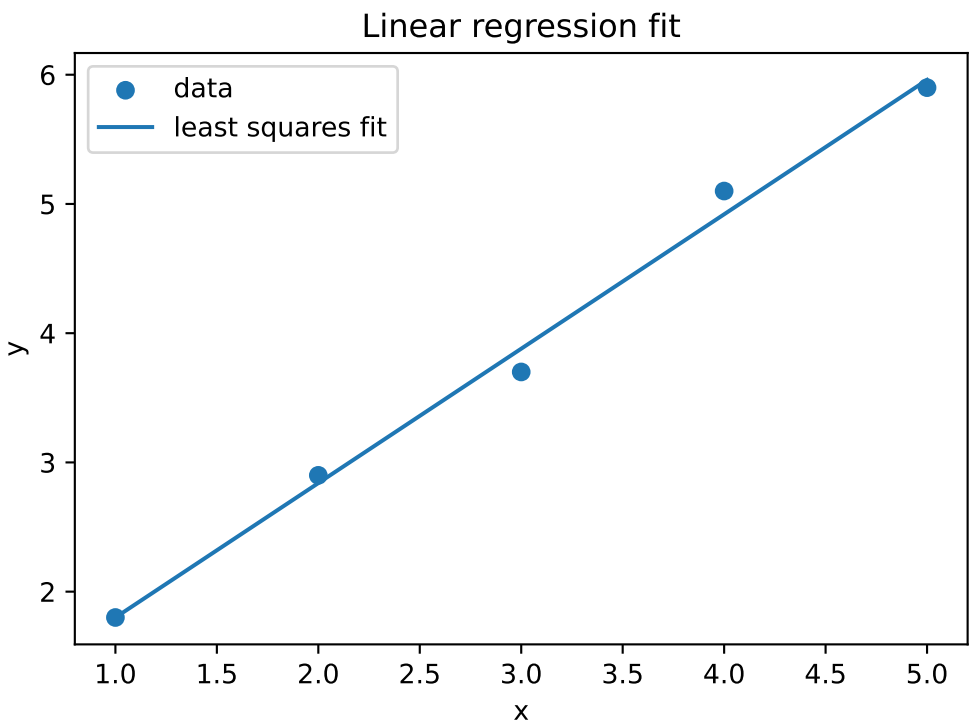
23.5 8.5 Predictions and residuals

```python
y_pred = X @ w
residuals = y - y_pred

print("predictions =", y_pred)
print("residuals =", residuals)
print("MSE =", np.mean(residuals**2))
```

```
predictions = [1.8  2.84 3.88 4.92 5.96]
residuals = [-2.22044605e-16  6.00000000e-02 -1.80000000e-01  1.80000000e-01
 -6.00000000e-02]
MSE = 0.014399999999999918
```

23.6 8.6 Visualization

```
plt.figure(figsize=(6, 4))
plt.scatter(X_raw, y, label="data")
plt.plot(X_raw, y_pred, label="least squares fit")
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.title("Linear regression fit")
plt.show()
```



23.7 8.7 Geometric interpretation

The normal equation

$$X^\top(Xw^* - y) = 0$$

says that the residual vector $r = y - Xw^*$ is orthogonal to the column space of X. Thus Xw^* is the orthogonal projection of y onto the column space of X.

This is one of the deepest conceptual points in the course. Least squares is not merely an algorithm; it is an orthogonal projection problem.

23.8 8.8 A synthetic multi-feature example

```
rng = np.random.default_rng(0)

n = 200
d = 3

X_features = rng.normal(size=(n, d))
true_w = np.array([1.5, -2.0, 0.75])
noise = 0.3 * rng.standard_normal(n)
y = X_features @ true_w + noise

X = X_features
w_hat = np.linalg.solve(X.T @ X, X.T @ y)

print("true_w =", true_w)
print("w_hat  =", w_hat)
print("estimation error =", np.linalg.norm(w_hat - true_w))
```

```
true_w = [ 1.5  -2.    0.75]
w_hat  = [ 1.47371393 -2.02925259  0.76647239]
estimation error = 0.04263814767727305
```

23.9 8.9 Regression with an explicit intercept

```
X_aug = np.column_stack([np.ones(n), X_features])
w_aug = np.linalg.solve(X_aug.T @ X_aug, X_aug.T @ y)

print("w_aug =", w_aug)
```

```
w_aug = [-0.01084165  1.47300522 -2.02884578  0.76590767]
```

The first component is the intercept.

23.10 8.10 Exercises

1. Derive the normal equation carefully starting from $f(w) = \|Xw - y\|_2^2$.
2. Implement least squares on a small one-dimensional dataset.
3. Add noise to the outputs and observe how the solution changes.
4. Compare the true coefficient vector to the estimated one on synthetic data.
5. Explain the projection interpretation in your own words.

23.11 8.11 Worked solution example

```
rng = np.random.default_rng(5)

n = 80
x = np.linspace(-2, 2, n)
y = 1.2 + 2.7 * x + 0.5 * rng.standard_normal(n)

X = np.column_stack([np.ones(n), x])
w = np.linalg.solve(X.T @ X, X.T @ y)
y_pred = X @ w

print("estimated parameters =", w)
print("MSE =", np.mean((y - y_pred)**2))

plt.figure(figsize=(6, 4))
plt.scatter(x, y, s=18)
plt.plot(x, y_pred)
plt.title("Worked least squares example")
plt.show()
```

```
estimated parameters = [1.11582782 2.73530772]
MSE = 0.20363506927607586
```

Worked least squares example
6
4
2
0
−2
−4
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

24 9. Gradient Descent for Least Squares

24.1 9.1 Why gradient descent?

The normal equation is elegant, but direct linear solves are not the only way to optimize an objective. Machine learning often relies on iterative optimization.

For least squares we consider the empirical risk

$$f(w) = \frac{1}{n}\|Xw - y\|_2^2.$$

Its gradient is

$$\nabla f(w) = \frac{2}{n}X^\top(Xw - y).$$

The gradient descent iteration is

$$w_{k+1} = w_k - \eta \nabla f(w_k),$$

where $\eta > 0$ is the learning rate.

24.2 9.2 Implementation

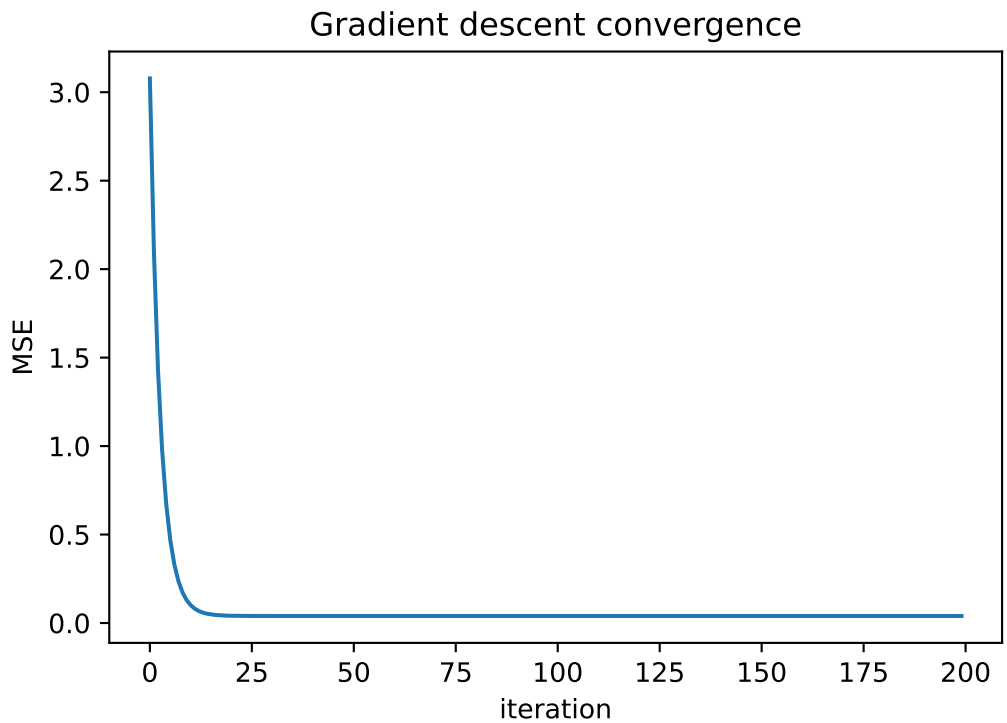
```
rng = np.random.default_rng(0)

n = 120
d = 2
X = rng.normal(size=(n, d))
true_w = np.array([2.0, -1.0])
y = X @ true_w + 0.2 * rng.standard_normal(n)

w = np.zeros(d)
```

```
eta = 0.1
num_steps = 200

loss_history = []

for _ in range(num_steps):
    grad = (2 / n) * X.T @ (X @ w - y)
    w = w - eta * grad
    loss = np.mean((X @ w - y)**2)
    loss_history.append(loss)

print("estimated w =", w)
print("true_w =", true_w)
print("final loss =", loss_history[-1])
```

```
estimated w = [ 1.99927294 -1.03094039]
true_w = [ 2. -1.]
final loss = 0.039563151213584634
```

24.3 9.3 Convergence plot

```
plt.figure(figsize=(6, 4))
plt.plot(loss_history)
plt.xlabel("iteration")
plt.ylabel("MSE")
plt.title("Gradient descent convergence")
plt.show()
```



24.4 9.4 Comparison with closed-form least squares

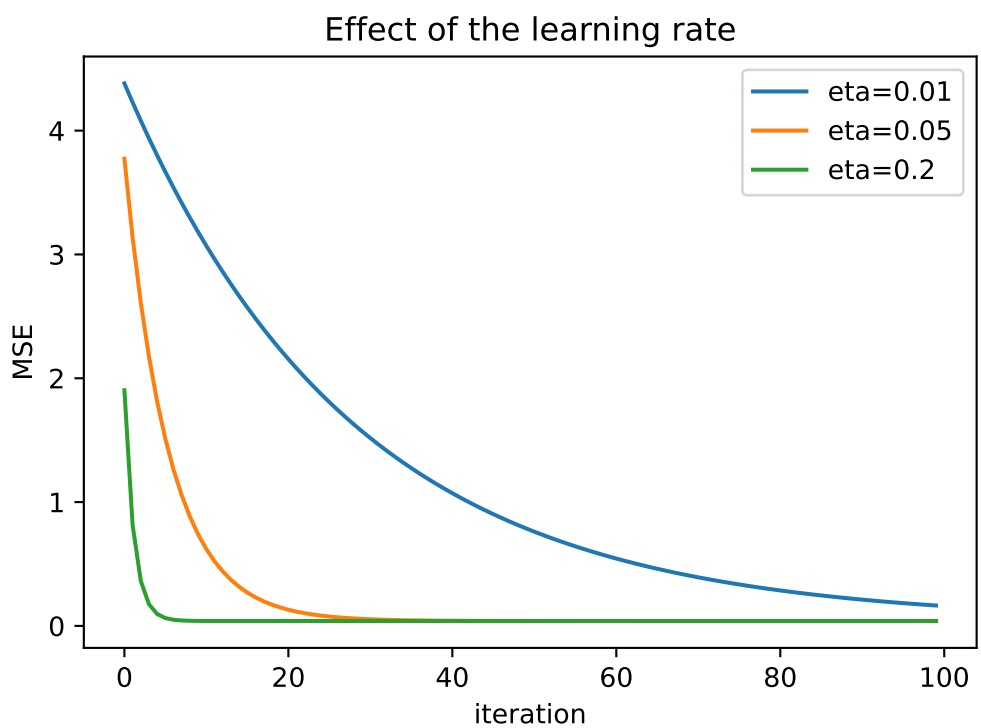
```
w_closed = np.linalg.solve(X.T @ X, X.T @ y)

print("w_closed =", w_closed)
print("w_gd     =", w)
print("difference =", np.linalg.norm(w_closed - w))
```

```
w_closed = [ 1.99927294 -1.03094039]
w_gd     = [ 1.99927294 -1.03094039]
difference = 9.42055475210265e-16
```

24.5 9.5 Learning rate effects

The choice of learning rate matters.

```
etas = [0.01, 0.05, 0.2]
histories = []

for eta in etas:
    w_eta = np.zeros(d)
    hist = []
    for _ in range(100):
        grad = (2 / n) * X.T @ (X @ w_eta - y)
        w_eta -= eta * grad
        hist.append(np.mean((X @ w_eta - y)**2))
    histories.append(hist)

plt.figure(figsize=(6, 4))
for eta, hist in zip(etas, histories):
    plt.plot(hist, label=f"eta={eta}")
plt.legend()
plt.xlabel("iteration")
plt.ylabel("MSE")
plt.title("Effect of the learning rate")
plt.show()
```



24.6 9.6 Gradient checking idea

A very useful debugging principle is to compare an analytic gradient with a finite-difference approximation.

For a scalar function $f : \mathbb{R}^d \to \mathbb{R}$ and a small perturbation h, one can compare

$$\frac{f(w + he_j) - f(w - he_j)}{2h}$$

to the j-th component of the analytic gradient.

```
def least_squares_loss(w, X, y):
    return np.mean((X @ w - y)**2)

def least_squares_grad(w, X, y):
    n = len(y)
    return (2 / n) * X.T @ (X @ w - y)

w0 = np.array([0.3, -0.2])
g = least_squares_grad(w0, X, y)

h = 1e-6
g_fd = np.zeros_like(w0)
for j in range(len(w0)):
    e = np.zeros_like(w0)
    e[j] = 1.0
    g_fd[j] = (
        least_squares_loss(w0 + h * e, X, y)
        - least_squares_loss(w0 - h * e, X, y)
    ) / (2 * h)

print("analytic grad =", g)
print("finite diff grad =", g_fd)
print("difference =", np.linalg.norm(g - g_fd))
```

```
analytic grad = [-3.12182606  1.2955987 ]
finite diff grad = [-3.12182606  1.2955987 ]
difference = 1.9233228267753716e-10
```

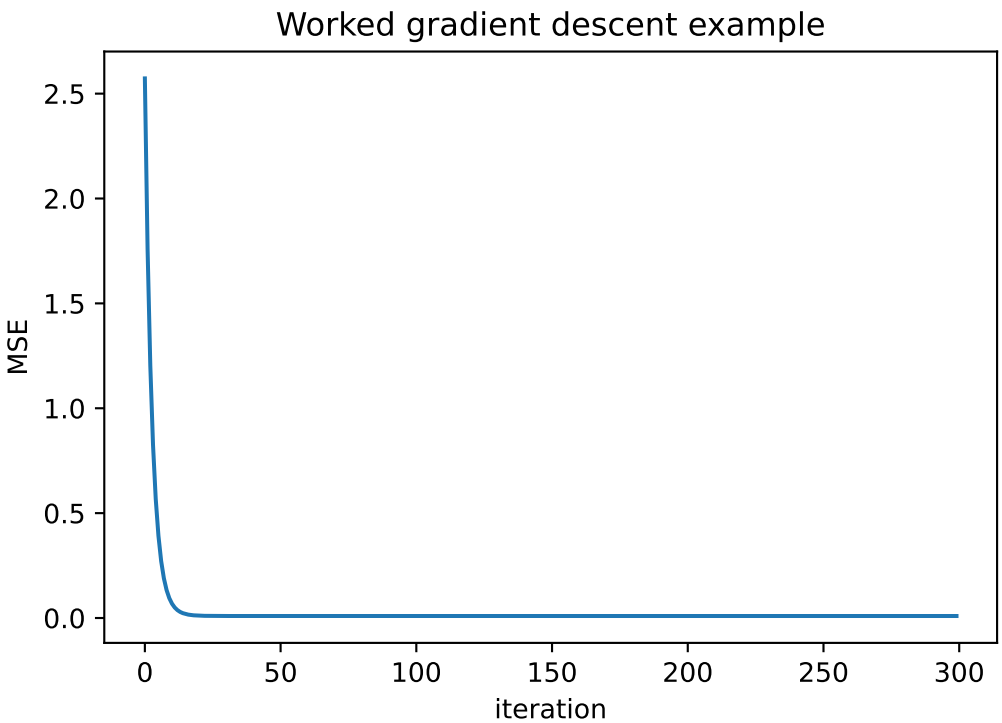
24.7 9.7 Exercises

1. Derive the gradient of the least squares loss.
2. Implement gradient descent for a synthetic regression problem.
3. Compare the result to the normal equation solution.
4. Experiment with too-small and too-large learning rates.
5. Use a finite-difference test to validate your gradient formula.

24.8 9.8 Worked example

```
rng = np.random.default_rng(11)

n = 150
d = 3
X = rng.normal(size=(n, d))
w_true = np.array([1.0, -1.5, 0.5])
y = X @ w_true + 0.1 * rng.normal(size=n)

w = np.zeros(d)
eta = 0.08
losses = []

for _ in range(300):
    grad = (2 / n) * X.T @ (X @ w - y)
    w -= eta * grad
    losses.append(np.mean((X @ w - y)**2))

print("w_true =", w_true)
print("w =", w)

plt.figure(figsize=(6, 4))
plt.plot(losses)
plt.title("Worked gradient descent example")
plt.xlabel("iteration")
plt.ylabel("MSE")
plt.show()
```

```
w_true = [ 1.  -1.5  0.5]
w = [ 0.9990376  -1.49071693  0.49701198]
```


Worked gradient descent example
MSE
iteration
2.5
2.0
1.5
1.0
0.5
0.0
0
50
100
150
200
250
300

25 10. Binary Classification and Logistic Regression

25.1 10.1 Why linear regression is not enough for classification

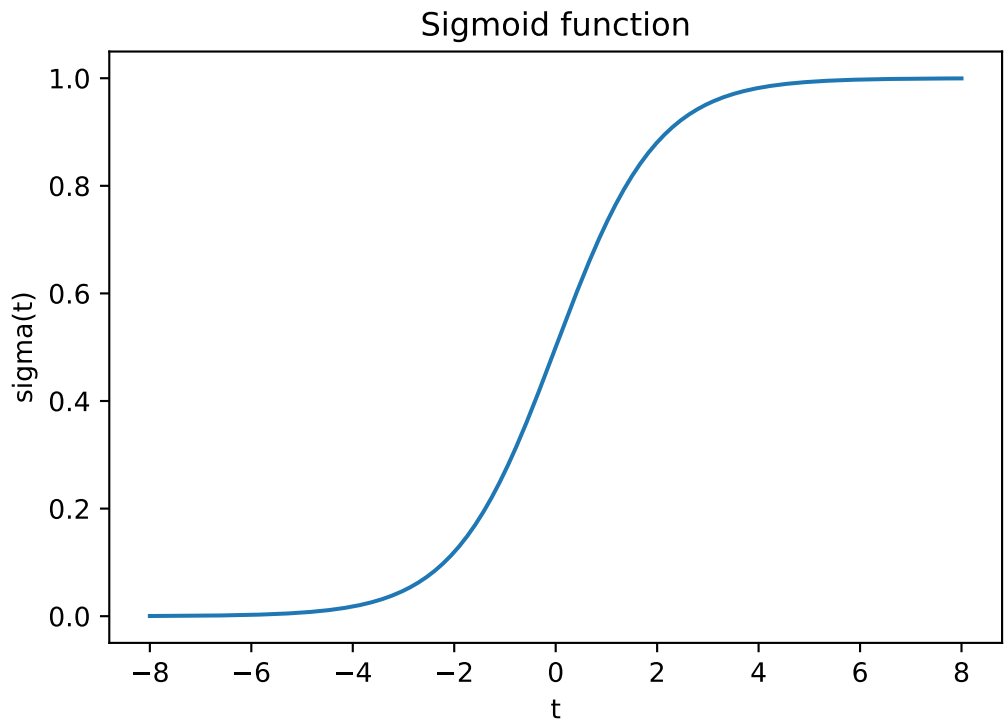
If the target values are 0 and 1, then least squares can still be used, but it does not model probabilities naturally and can produce values outside the interval $[0, 1]$.

To model probabilities we use the sigmoid function

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

25.2 10.2 The sigmoid function

```
def sigmoid(t):
    return 1 / (1 + np.exp(-t))

t = np.linspace(-8, 8, 400)

plt.figure(figsize=(6, 4))
plt.plot(t, sigmoid(t))
plt.xlabel("t")
plt.ylabel("sigma(t)")
plt.title("Sigmoid function")
plt.show()
```



25.3 10.3 Logistic model

Given $x \in \mathbb{R}^d$ and parameter vector $w \in \mathbb{R}^d$, the logistic model is

$$p(y = 1 \mid x) = \sigma(x^\top w).$$

For a full dataset with design matrix X, the vector of predicted probabilities is

$$p = \sigma(Xw),$$

where the sigmoid is applied componentwise.

25.4 10.4 Binary cross-entropy loss

If $y_i \in \{0, 1\}$ and p_i is the predicted probability of class 1, the average logistic loss is

$$L(w) = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log p_i + (1-y_i)\log(1-p_i)\right).$$

This is also called binary cross-entropy.

25.5 10.5 Gradient of the logistic loss

A standard calculation shows that

$$\nabla L(w) = \frac{1}{n}X^\top(p-y),$$

where

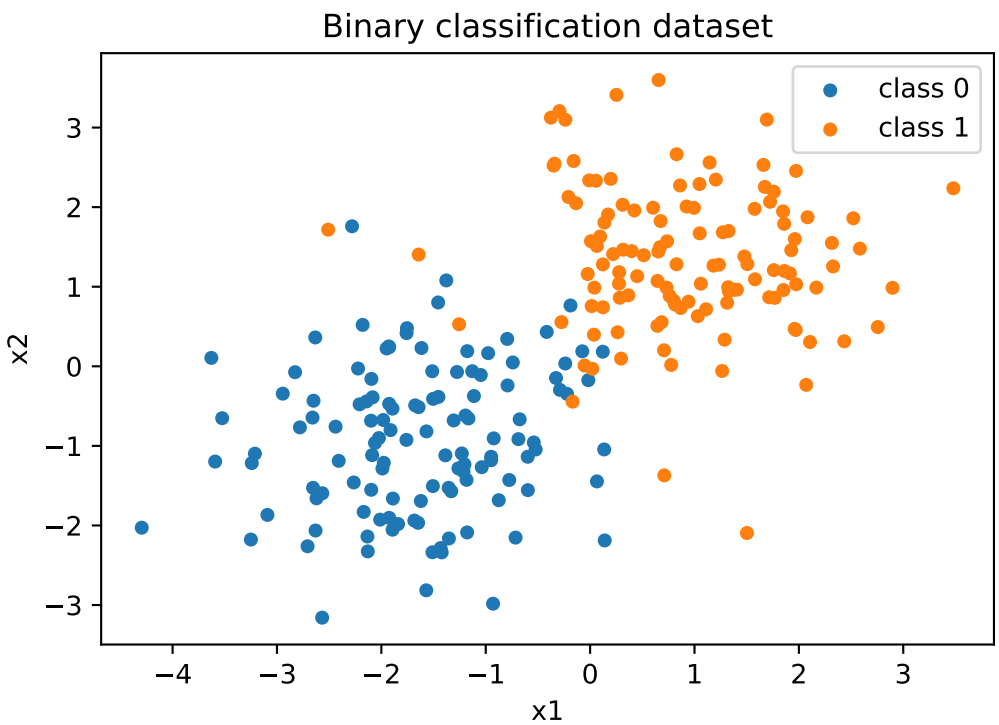
$$p = \sigma(Xw).$$

This compact formula is one of the main reasons logistic regression is such an elegant teaching example.

25.6 10.6 Synthetic dataset

```
rng = np.random.default_rng(0)

n0 = 120
n1 = 120

X0 = rng.normal(loc=[-1.5, -1.0], scale=0.9, size=(n0, 2))
X1 = rng.normal(loc=[1.0, 1.3], scale=0.9, size=(n1, 2))

X_raw = np.vstack([X0, X1])
y = np.concatenate([np.zeros(n0), np.ones(n1)])

plt.figure(figsize=(6, 4))
plt.scatter(X0[:, 0], X0[:, 1], s=18, label="class 0")
plt.scatter(X1[:, 0], X1[:, 1], s=18, label="class 1")
plt.legend()
plt.xlabel("x1")
plt.ylabel("x2")
```

```
plt.title("Binary classification dataset")
plt.show()
```



25.7 10.7 Add the intercept column

```
X = np.column_stack([np.ones(len(X_raw)), X_raw])
print("X.shape =", X.shape)
print("y.shape =", y.shape)
```

```
X.shape = (240, 3)
y.shape = (240,)
```

25.8 10.8 Binary cross-entropy implementation

```
def binary_cross_entropy(y_true, y_prob):
    eps = 1e-12
    y_prob = np.clip(y_prob, eps, 1 - eps)
    return -np.mean(
        y_true * np.log(y_prob)
        + (1 - y_true) * np.log(1 - y_prob)
    )
```

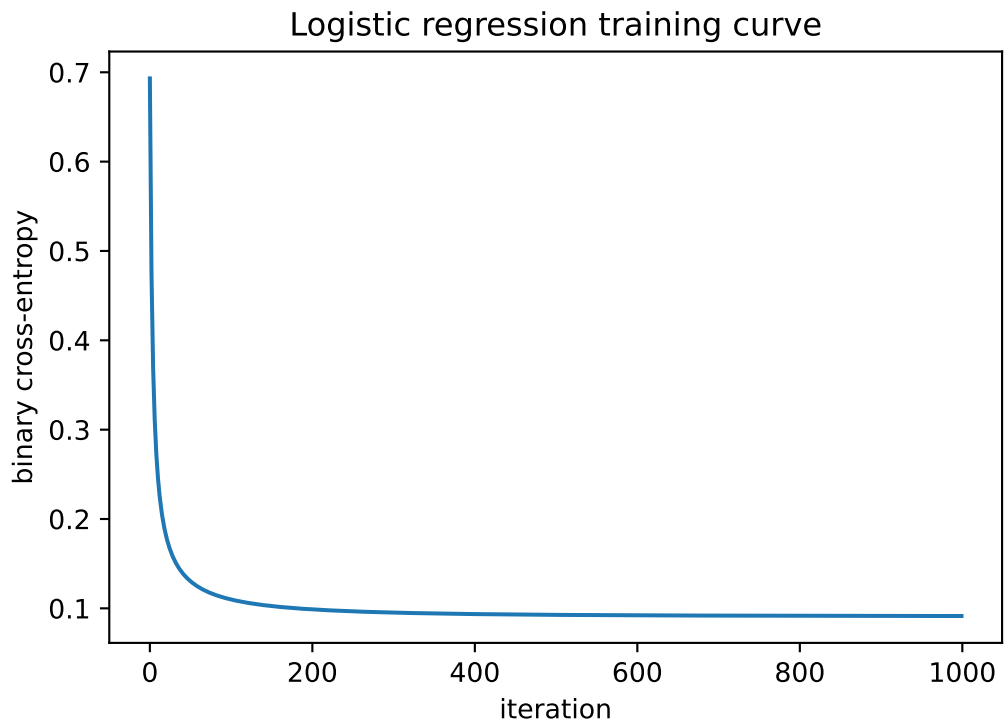
25.9 10.9 Gradient descent for logistic regression

```
n, d = X.shape
w = np.zeros(d)
eta = 0.2
num_steps = 1000

history = []

for _ in range(num_steps):
    p = sigmoid(X @ w)
    grad = (X.T @ (p - y)) / n
    w -= eta * grad
    history.append(binary_cross_entropy(y, p))

print("w =", w)
print("final loss =", history[-1])
```

```
w = [0.40651878 2.72978342 2.35158575]
final loss = 0.09141766952958857
```

25.10 10.10 Loss curve

```
plt.figure(figsize=(6, 4))
plt.plot(history)
plt.xlabel("iteration")
plt.ylabel("binary cross-entropy")
plt.title("Logistic regression training curve")
plt.show()
```



25.11 10.11 Classification accuracy

```
p = sigmoid(X @ w)
y_pred = (p >= 0.5).astype(int)

accuracy = np.mean(y_pred == y)
print("accuracy =", accuracy)
```

```
accuracy = 0.9583333333333334
```

25.12 10.12 Decision boundary visualization

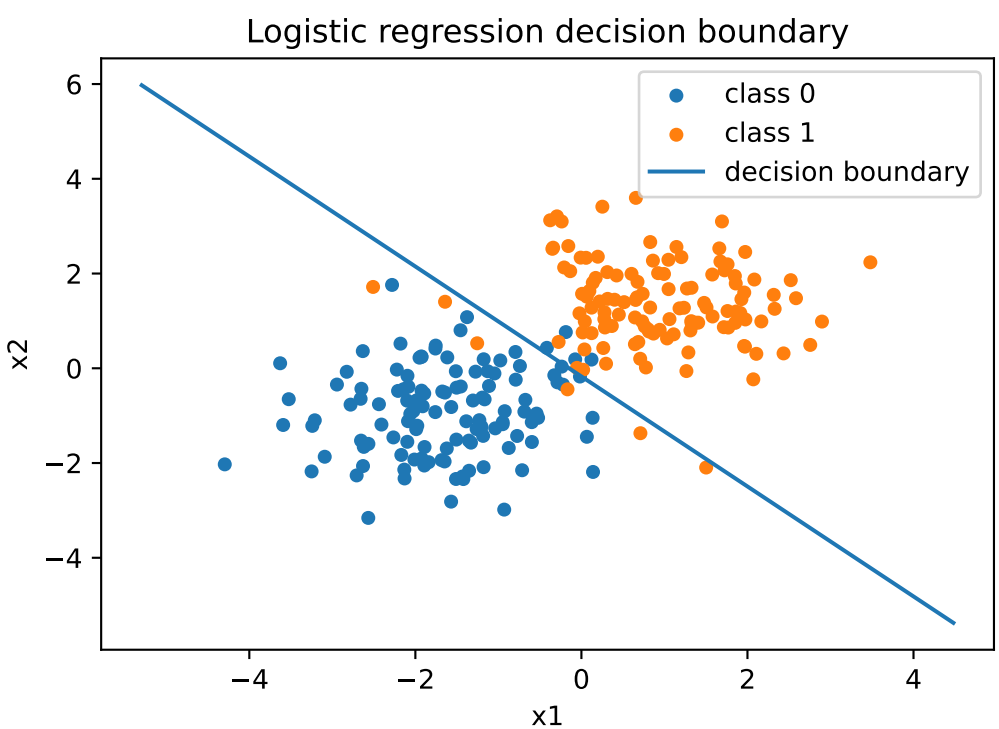
For a model

$$p(y = 1 \mid x) = \sigma(w_0 + w_1 x_1 + w_2 x_2),$$

the decision boundary at threshold 0.5 is

$$w_0 + w_1 x_1 + w_2 x_2 = 0.$$

```
plt.figure(figsize=(6, 4))
plt.scatter(X0[:, 0], X0[:, 1], s=18, label="class 0")
plt.scatter(X1[:, 0], X1[:, 1], s=18, label="class 1")

x1_min, x1_max = X_raw[:, 0].min() - 1, X_raw[:, 0].max() + 1
grid_x1 = np.linspace(x1_min, x1_max, 200)

if abs(w[2]) > 1e-12:
    grid_x2 = -(w[0] + w[1] * grid_x1) / w[2]
    plt.plot(grid_x1, grid_x2, label="decision boundary")

plt.xlabel("x1")
plt.ylabel("x2")
plt.legend()
plt.title("Logistic regression decision boundary")
plt.show()
```



25.13 10.13 Confusion counts

```
tp = np.sum((y_pred == 1) & (y == 1))
tn = np.sum((y_pred == 0) & (y == 0))
fp = np.sum((y_pred == 1) & (y == 0))
fn = np.sum((y_pred == 0) & (y == 1))

print("TP =", tp)
print("TN =", tn)
print("FP =", fp)
print("FN =", fn)
```

```
TP = 114
TN = 116
FP = 4
FN = 6
```

25.14 10.14 Exercises

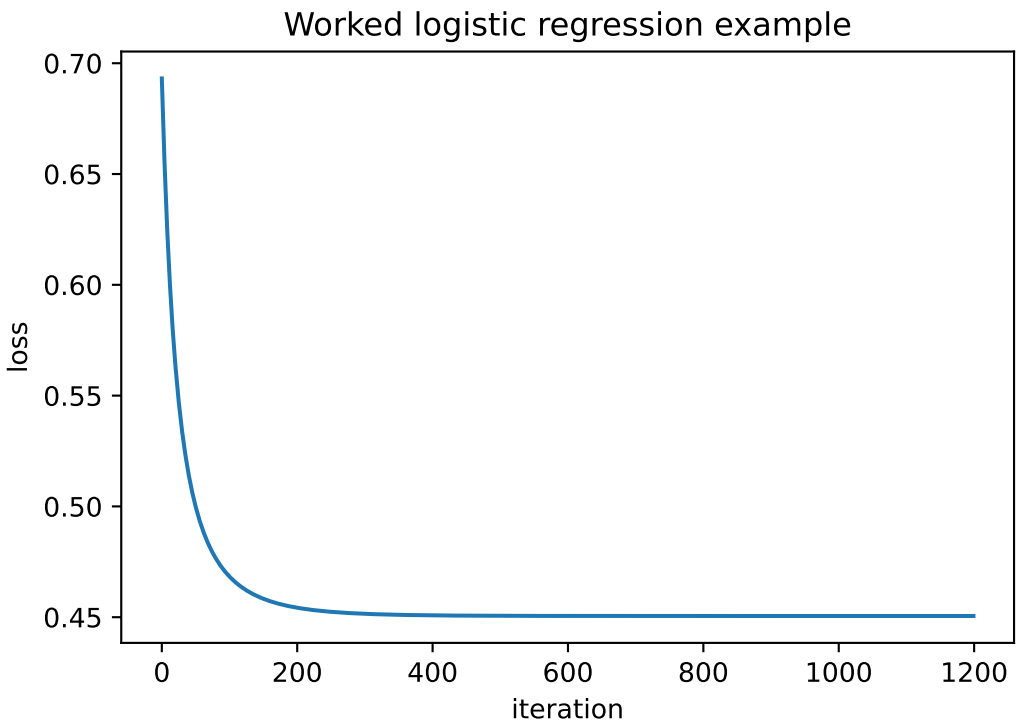
1. Derive the logistic regression gradient formula.
2. Train logistic regression on a synthetic 2D dataset.
3. Plot the decision boundary.
4. Compute the accuracy and confusion counts.
5. Explain why clipping probabilities is useful in the cross-entropy implementation.

25.15 10.15 Worked example

```
rng = np.random.default_rng(17)

n = 200
X_raw = rng.normal(size=(n, 2))
true_w = np.array([-0.5, 2.0, -1.2])

X = np.column_stack([np.ones(n), X_raw])
p_true = sigmoid(X @ true_w)
y = (rng.random(n) < p_true).astype(float)
```

```python
w = np.zeros(X.shape[1])
eta = 0.1
losses = []

for _ in range(1200):
    p = sigmoid(X @ w)
    grad = X.T @ (p - y) / n
    w -= eta * grad
    losses.append(binary_cross_entropy(y, p))

print("true_w =", true_w)
print("estimated_w =", w)
print("accuracy =", np.mean((sigmoid(X @ w) >= 0.5) == y))

plt.figure(figsize=(6, 4))
plt.plot(losses)
plt.title("Worked logistic regression example")
plt.xlabel("iteration")
plt.ylabel("loss")
plt.show()
```

```
true_w = [-0.5  2.  -1.2]
estimated_w = [-0.36513097  1.69627472 -1.06355111]
accuracy = 0.78
```


Worked logistic regression example
0.70
0.65
0.60
0.55
0.50
0.45
loss
0
200
400
600
800
1000
1200
iteration

26 11. Data Preprocessing, Features, and Shapes

26.1 11.1 Why preprocessing matters

A model is not only determined by its optimization method. The representation of the data matters.

Common preprocessing tasks include:

- selecting relevant columns;
- handling missing values;
- encoding categorical information;
- centering and scaling;
- constructing interaction features;
- splitting data into training and testing sets.

26.2 11.2 Reshape, flatten, and new axes

```
x = np.arange(12)
X = x.reshape(3, 4)

print("x =", x)
print("X =")
print(X)
print("x.shape =", x.shape)
print("X.shape =", X.shape)
print("flatten =", X.flatten())
print("ravel   =", X.ravel())
print("x[:, None].shape =", np.array([1, 2, 3])[:, None].shape)
```

```
x = [ 0  1  2  3  4  5  6  7  8  9 10 11]
X =
```

```
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
x.shape = (12,)
X.shape = (3, 4)
flatten = [ 0  1  2  3  4  5  6  7  8  9 10 11]
ravel   = [ 0  1  2  3  4  5  6  7  8  9 10 11]
x[:, None].shape = (3, 1)
```

26.3 11.3 Feature scaling

```python
rng = np.random.default_rng(9)

X = np.column_stack([
    rng.normal(0, 1, size=200),
    rng.normal(1000, 250, size=200),
    rng.normal(0.01, 0.003, size=200)
])

mu = np.mean(X, axis=0)
sigma = np.std(X, axis=0)
Z = (X - mu) / sigma

print("original feature scales:")
print(np.std(X, axis=0))
print("scaled feature scales:")
print(np.std(Z, axis=0))
```

```
original feature scales:
[1.02119562e+00 2.42888357e+02 3.16507367e-03]
scaled feature scales:
[1. 1. 1.]
```

Scaling is often crucial for gradient-based optimization because poorly scaled features can slow convergence.

26.4 11.4 Train-test split using pandas

```
df = df_csv.copy()

numeric_cols = [c for c in df.columns if pd.api.types.is_numeric_dtype(df[c])]
target_col = "target" if "target" in df.columns else numeric_cols[-1]
feature_cols = [c for c in numeric_cols if c != target_col]

df_model = df[feature_cols + [target_col]].dropna()

X = df_model[feature_cols].to_numpy(dtype=float)
y = df_model[target_col].to_numpy(dtype=float)

rng = np.random.default_rng(123)
perm = rng.permutation(len(X))
split = int(0.8 * len(X))

X_train = X[perm[:split]]
X_test = X[perm[split:]]
y_train = y[perm[:split]]
y_test = y[perm[split:]]

print("feature columns =", feature_cols)
print("X_train.shape =", X_train.shape)
print("X_test.shape =", X_test.shape)
```

```
feature columns = ['age', 'income', 'hours_studied']
X_train.shape = (96, 3)
X_test.shape = (24, 3)
```

26.5 11.5 Polynomial features in one variable

A linear model becomes nonlinear in the original input if we augment the feature map.

For one variable x, define

$$\phi(x) = (1, x, x^2, x^3).$$

Then the model

$$\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3$$

is still linear in the parameter vector w.

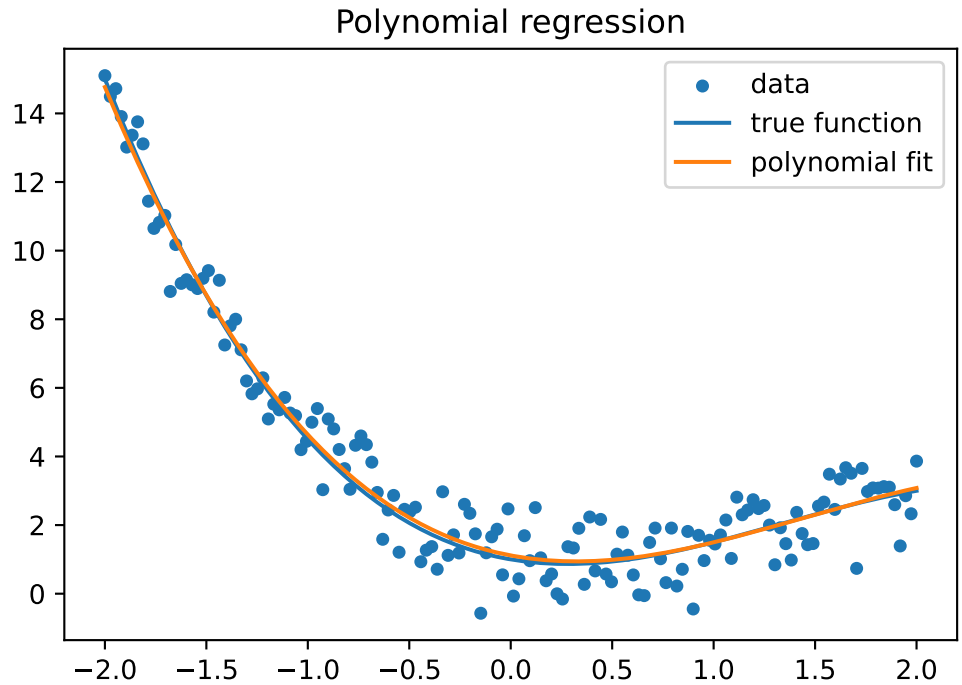
```
x = np.linspace(-2, 2, 100)
Phi = np.column_stack([np.ones_like(x), x, x**2, x**3])

print("Phi.shape =", Phi.shape)
print(Phi[:5])
```

```
Phi.shape = (100, 4)
[[ 1.         -2.          4.         -8.        ]
 [ 1.         -1.95959596  3.84001632 -7.52488047]
 [ 1.         -1.91919192  3.68329762 -7.06895503]
 [ 1.         -1.87878788  3.52984389 -6.63182792]
 [ 1.         -1.83838384  3.37965514 -6.21310338]]
```

26.6 11.6 A polynomial regression example

```
rng = np.random.default_rng(0)

x = np.linspace(-2, 2, 150)
y_true = 1 - x + 2 * x**2 - 0.5 * x**3
y = y_true + 0.8 * rng.standard_normal(len(x))

Phi = np.column_stack([np.ones_like(x), x, x**2, x**3])
w = np.linalg.solve(Phi.T @ Phi, Phi.T @ y)
y_pred = Phi @ w

print("estimated coefficients =", w)

plt.figure(figsize=(6, 4))
plt.scatter(x, y, s=15, label="data")
plt.plot(x, y_true, label="true function")
plt.plot(x, y_pred, label="polynomial fit")
plt.legend()
plt.title("Polynomial regression")
plt.show()
```

```
estimated coefficients = [ 1.11483741 -1.11943614  1.95232421 -0.45054691]
```



26.7 11.7 Exercises

1. Explain the difference between `reshape`, `flatten`, and `ravel`.
2. Standardize a feature matrix and verify columnwise mean 0 and standard deviation 1.
3. Build polynomial features up to degree 3 for a one-dimensional dataset.
4. Split a dataset into train and test parts.
5. Explain why feature engineering can make a linear model nonlinear in the original input.

26.8 11.8 Worked example

```
x = np.linspace(-1, 1, 8)
Phi = np.column_stack([np.ones_like(x), x, x**2])
print("Phi =")
print(Phi)
```

```python
mu = np.mean(Phi[:, 1:], axis=0)
sigma = np.std(Phi[:, 1:], axis=0)
Phi_scaled = Phi.copy()
Phi_scaled[:, 1:] = (Phi_scaled[:, 1:] - mu) / sigma

print()
print("Phi_scaled =")
print(Phi_scaled)
```

```
Phi =
[[ 1.         -1.          1.        ]
 [ 1.         -0.71428571  0.51020408]
 [ 1.         -0.42857143  0.18367347]
 [ 1.         -0.14285714  0.02040816]
 [ 1.          0.14285714  0.02040816]
 [ 1.          0.42857143  0.18367347]
 [ 1.          0.71428571  0.51020408]
 [ 1.          1.          1.        ]]

Phi_scaled =
[[ 1.         -1.52752523  1.52752523]
 [ 1.         -1.09108945  0.21821789]
 [ 1.         -0.65465367 -0.65465367]
 [ 1.         -0.21821789 -1.09108945]
 [ 1.          0.21821789 -1.09108945]
 [ 1.          0.65465367 -0.65465367]
 [ 1.          1.09108945  0.21821789]
 [ 1.          1.52752523  1.52752523]]
```

27 12. Regularization and Overfitting

27.1 12.1 Motivation

When a model is too flexible relative to the amount of data, it may fit noise rather than structure. This phenomenon is called overfitting.

One way to control complexity is regularization.

27.2 12.2 Ridge regression

The ridge objective is

$$f_\lambda(w) = \|Xw - y\|_2^2 + \lambda\|w\|_2^2$$

for $\lambda > 0$.

Differentiating gives

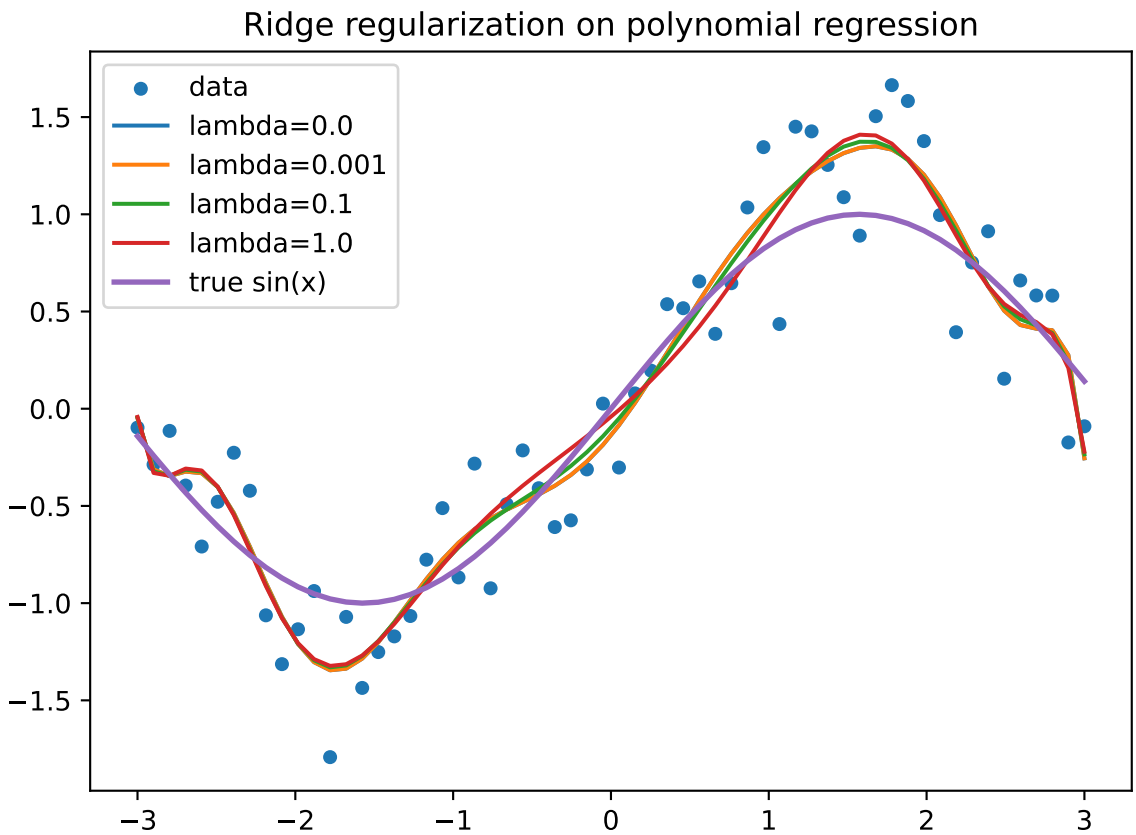
$$\nabla f_\lambda(w) = 2X^\top(Xw - y) + 2\lambda w.$$

The normal equation becomes

$$(X^\top X + \lambda I)w = X^\top y.$$

27.3 12.3 Ridge implementation

```python
rng = np.random.default_rng(0)

n = 60
x = np.linspace(-3, 3, n)
y_true = np.sin(x)
y = y_true + 0.35 * rng.standard_normal(n)

degree = 12
Phi = np.column_stack([x**k for k in range(degree + 1)])

lam_values = [0.0, 1e-3, 1e-1, 1.0]

plt.figure(figsize=(7, 5))
plt.scatter(x, y, s=18, label="data")

for lam in lam_values:
    A = Phi.T @ Phi + lam * np.eye(Phi.shape[1])
    w = np.linalg.solve(A, Phi.T @ y)
    y_pred = Phi @ w
    plt.plot(x, y_pred, label=f"lambda={lam}")

plt.plot(x, y_true, label="true sin(x)", linewidth=2)
plt.legend()
plt.title("Ridge regularization on polynomial regression")
plt.show()
```



27.4 12.4 Train-test perspective

A model can have very small training error and much worse test error. That discrepancy is a practical sign of overfitting.

```
rng = np.random.default_rng(1)

perm = rng.permutation(n)
split = int(0.7 * n)
train_idx = perm[:split]
test_idx = perm[split:]

x_train = x[train_idx]
y_train = y[train_idx]
x_test = x[test_idx]
y_test = y[test_idx]
```

```
Phi_train = np.column_stack([x_train**k for k in range(degree + 1)])
Phi_test = np.column_stack([x_test**k for k in range(degree + 1)])

for lam in [0.0, 1e-3, 1e-2, 1e-1, 1.0]:
    A = Phi_train.T @ Phi_train + lam * np.eye(Phi_train.shape[1])
    w = np.linalg.solve(A, Phi_train.T @ y_train)
    train_mse = np.mean((Phi_train @ w - y_train)**2)
    test_mse = np.mean((Phi_test @ w - y_test)**2)
    print(f"lambda={lam:>5}: train_mse={train_mse:.4f}, test_mse={test_mse:.4f}")
```

```
lambda=  0.0: train_mse=0.0551, test_mse=0.1117
lambda=0.001: train_mse=0.0551, test_mse=0.1116
lambda= 0.01: train_mse=0.0551, test_mse=0.1113
lambda=  0.1: train_mse=0.0563, test_mse=0.1108
lambda=  1.0: train_mse=0.0613, test_mse=0.1198
```

27.5 12.5 Exercises

1. Derive the ridge normal equation.
2. Fit a high-degree polynomial with and without regularization.
3. Compare train and test errors for different values of λ.
4. Explain why ridge can improve generalization.
5. Discuss whether the intercept should always be regularized in practical applications.

27.6 12.6 Worked example

```
rng = np.random.default_rng(2)

x = np.linspace(-2, 2, 50)
y = np.cos(2 * x) + 0.2 * rng.normal(size=len(x))
Phi = np.column_stack([x**k for k in range(9)])

for lam in [0.0, 0.01, 0.1]:
    w = np.linalg.solve(Phi.T @ Phi + lam * np.eye(Phi.shape[1]), Phi.T @ y)
    print(f"lambda={lam}, coefficient norm={np.linalg.norm(w):.3f}")
```

```
lambda=0.0, coefficient norm=2.505
lambda=0.01, coefficient norm=2.391
lambda=0.1, coefficient norm=1.834
```

28 13. Mini-Batches and Stochastic Ideas

28.1 13.1 From full gradients to sampled gradients

For large datasets, computing the full gradient at every step can be expensive. Instead one can approximate it by sampling a subset of observations.

This leads to mini-batch gradient methods.

28.2 13.2 A simple batch sampler

```
rng = np.random.default_rng(0)

n = 500
d = 4
X = rng.normal(size=(n, d))
w_true = np.array([1.0, -2.0, 0.5, 1.5])
y = X @ w_true + 0.5 * rng.normal(size=n)

def sample_minibatch(X, y, batch_size, rng):
    idx = rng.choice(len(X), size=batch_size, replace=False)
    return X[idx], y[idx]

Xb, yb = sample_minibatch(X, y, batch_size=16, rng=rng)
print("Xb.shape =", Xb.shape)
print("yb.shape =", yb.shape)
```

```
Xb.shape = (16, 4)
yb.shape = (16,)
```

28.3 13.3 Mini-batch gradient descent for least squares

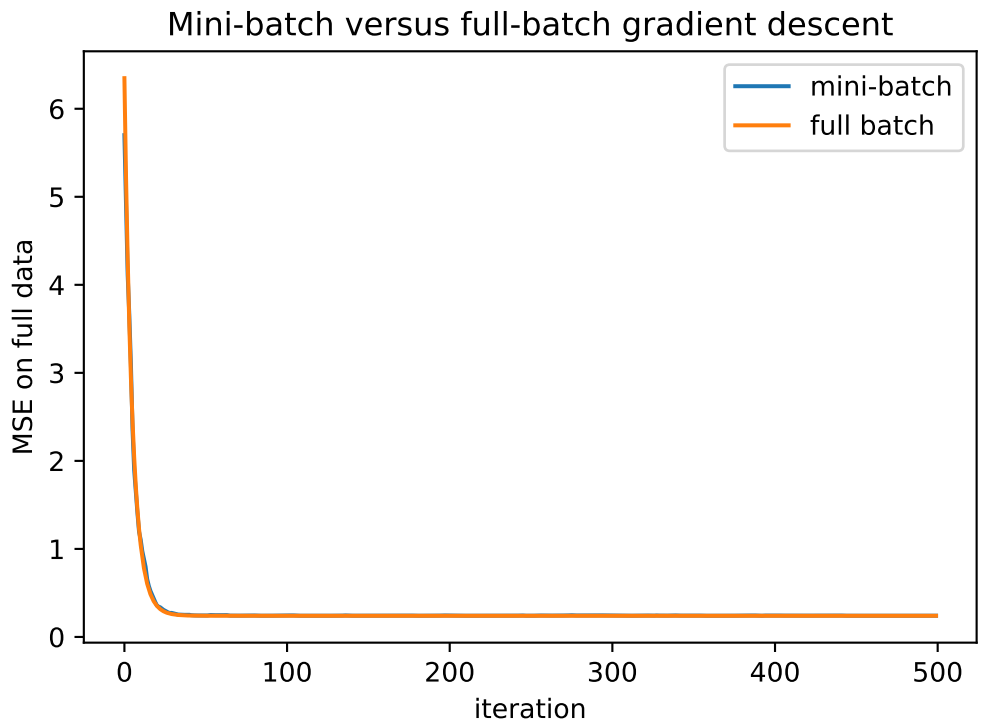
```
rng = np.random.default_rng(1)

w = np.zeros(d)
eta = 0.05
batch_size = 32
num_steps = 500

losses = []

for _ in range(num_steps):
    Xb, yb = sample_minibatch(X, y, batch_size=batch_size, rng=rng)
    grad = (2 / len(Xb)) * Xb.T @ (Xb @ w - yb)
    w -= eta * grad
    losses.append(np.mean((X @ w - y)**2))

print("estimated w =", w)
print("true w =", w_true)
```

```
estimated w = [ 1.00794574 -2.02964886  0.51013668  1.5059959 ]
true w = [ 1.  -2.   0.5  1.5]
```

28.4 13.4 Comparison with full-batch gradient descent

```
w_full = np.zeros(d)
losses_full = []

for _ in range(num_steps):
    grad_full = (2 / n) * X.T @ (X @ w_full - y)
    w_full -= eta * grad_full
    losses_full.append(np.mean((X @ w_full - y)**2))

plt.figure(figsize=(6, 4))
plt.plot(losses, label="mini-batch")
plt.plot(losses_full, label="full batch")
plt.legend()
plt.xlabel("iteration")
```

```
plt.ylabel("MSE on full data")
plt.title("Mini-batch versus full-batch gradient descent")
plt.show()
```



28.5 13.5 Exercises

1. Implement a mini-batch sampler.
2. Compare mini-batch and full-batch gradient descent on the same problem.
3. Change the batch size and observe the effect on the loss curve.
4. Explain why mini-batch methods introduce noise into the optimization path.
5. Discuss why stochastic methods are attractive for large-scale machine learning.

28.6 13.6 Worked example

```
rng = np.random.default_rng(4)
```

```python
n = 300
d = 2
X = rng.normal(size=(n, d))
w_true = np.array([2.5, -0.7])
y = X @ w_true + 0.3 * rng.normal(size=n)

def mbgd(X, y, batch_size, eta, steps, rng):
    w = np.zeros(X.shape[1])
    hist = []
    for _ in range(steps):
        idx = rng.choice(len(X), size=batch_size, replace=False)
        Xb = X[idx]
        yb = y[idx]
        grad = (2 / batch_size) * Xb.T @ (Xb @ w - yb)
        w -= eta * grad
        hist.append(np.mean((X @ w - y)**2))
    return w, hist

w_mb, hist_mb = mbgd(X, y, batch_size=24, eta=0.08, steps=250, rng=rng)
print("w_mb =", w_mb)
```

```
w_mb = [ 2.52243211 -0.67183942]
```

29 14. A First Neural Network in Pure NumPy

29.1 14.1 Why neural networks fit naturally into linear algebra

A fully connected neural network is built from repeated compositions of affine maps and nonlinear activation functions. This makes it an ideal topic after students are comfortable with matrices, broadcasting, and vectorization.

We consider a network with one hidden layer:

$$H = \phi(XW_1 + b_1),$$

$$\hat{Y} = HW_2 + b_2.$$

Here

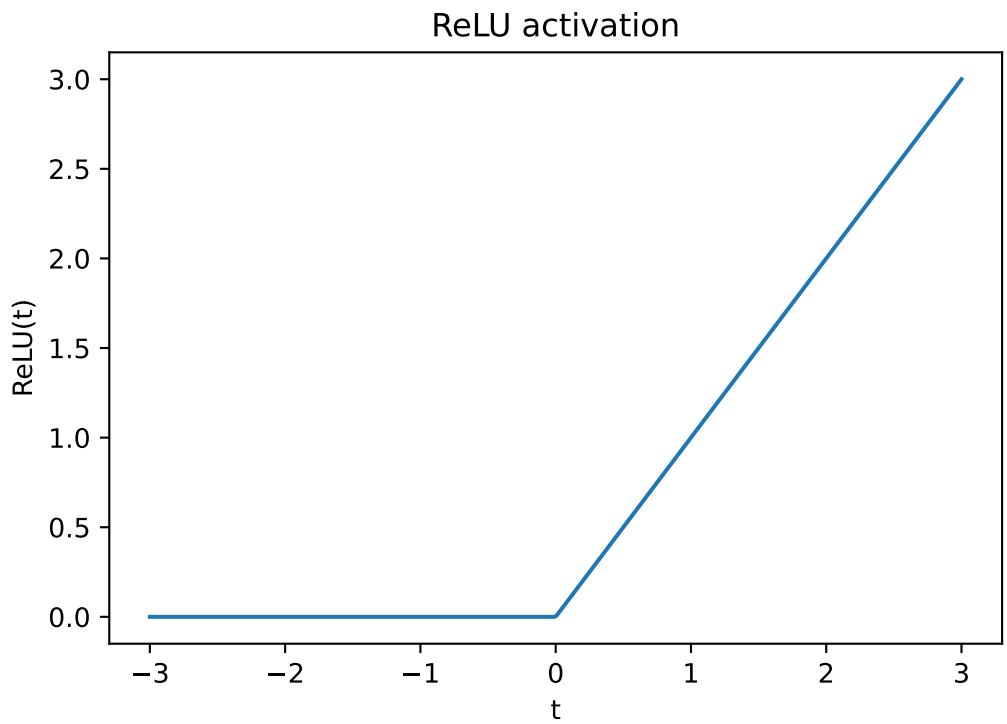
- $X \in \mathbb{R}^{n \times d}$ is the input matrix;
- $W_1 \in \mathbb{R}^{d \times h}$ is the first weight matrix;
- $b_1 \in \mathbb{R}^h$ is the hidden bias;
- $H \in \mathbb{R}^{n \times h}$ is the hidden representation;
- $W_2 \in \mathbb{R}^{h \times m}$ is the second weight matrix;
- $b_2 \in \mathbb{R}^m$ is the output bias.

29.2 14.2 Activation functions

We begin with the ReLU function

$$\mathrm{ReLU}(t) = \max(0, t).$$

```
def relu(x):
    return np.maximum(0, x)

t = np.linspace(-3, 3, 400)

plt.figure(figsize=(6, 4))
plt.plot(t, relu(t))
plt.xlabel("t")
plt.ylabel("ReLU(t)")
plt.title("ReLU activation")
plt.show()
```



29.3 14.3 Forward pass on XOR-style data

```
rng = np.random.default_rng(0)

X = np.array([
```

```
    [0.0, 0.0],
    [0.0, 1.0],
    [1.0, 0.0],
    [1.0, 1.0]
])

y = np.array([[0.0], [1.0], [1.0], [0.0]])

d = 2
h = 4
m = 1

W1 = 0.5 * rng.standard_normal((d, h))
b1 = np.zeros(h)
W2 = 0.5 * rng.standard_normal((h, m))
b2 = np.zeros(m)

H = relu(X @ W1 + b1)
Y = H @ W2 + b2

print("H =")
print(H)
print()
print("Y =")
print(Y)
```

```
H =
[[0.         0.         0.         0.        ]
 [0.         0.18079753 0.65200002 0.47354048]
 [0.06286511 0.         0.32021133 0.05245006]
 [0.         0.1147451  0.97221135 0.52599054]]

Y =
[[ 0.        ]
 [-0.30779526]
 [-0.12082619]
 [-0.36470917]]
```

29.4 14.4 Sigmoid output for binary classification

```
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

P = sigmoid(Y)
print("P =")
print(P)
```

```
P =
[[0.5       ]
 [0.42365298]
 [0.46983015]
 [0.40982009]]
```

29.5 14.5 Binary cross-entropy for network outputs

```
def bce_from_logits(logits, y_true):
    probs = sigmoid(logits)
    eps = 1e-12
    probs = np.clip(probs, eps, 1 - eps)
    return -np.mean(y_true * np.log(probs) + (1 - y_true) * np.log(1 - probs))
```

29.6 14.6 Backpropagation outline

For a first course, it is often enough to outline the structure of backpropagation without deriving every formula in maximal detail.

Let

$$Z_1 = XW_1 + b_1, \quad H = \mathrm{ReLU}(Z_1), \quad Z_2 = HW_2 + b_2, \quad P = \sigma(Z_2).$$

With binary cross-entropy and sigmoid output, the output error term is

$$\delta_2 = P - y.$$

Then the parameter gradients are

$$\nabla_{W_2} L = \frac{1}{n} H^\top \delta_2,$$

$$\nabla_{b_2} L = \frac{1}{n} \sum_{i=1}^{n} \delta_{2,i},$$

and for the hidden layer,

$$\delta_1 = (\delta_2 W_2^\top) \odot \mathbf{1}_{Z_1 > 0},$$

$$\nabla_{W_1} L = \frac{1}{n} X^\top \delta_1, \qquad \nabla_{b_1} L = \frac{1}{n} \sum_{i=1}^{n} \delta_1^{(i)}.$$

29.7 14.7 Training the network

```
rng = np.random.default_rng(1)

X = np.array([
    [0.0, 0.0],
    [0.0, 1.0],
    [1.0, 0.0],
    [1.0, 1.0]
])

y = np.array([[0.0], [1.0], [1.0], [0.0]])

d = 2
h = 8
m = 1

W1 = 0.5 * rng.standard_normal((d, h))
b1 = np.zeros(h)
W2 = 0.5 * rng.standard_normal((h, m))
b2 = np.zeros(m)

eta = 0.5
history = []
```

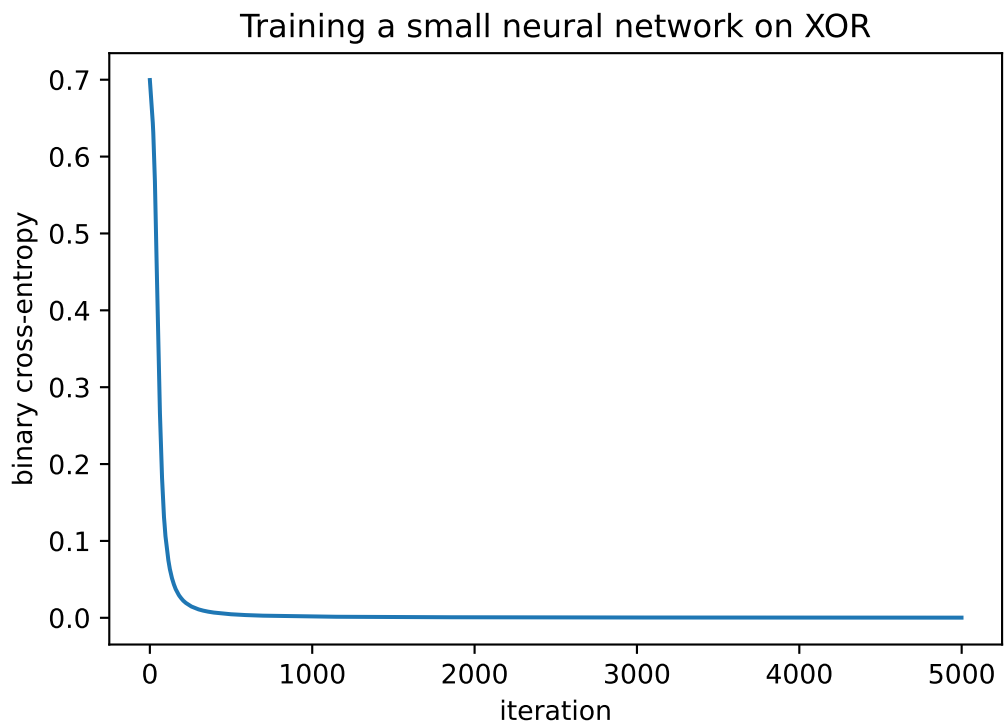
```
for _ in range(5000):
    Z1 = X @ W1 + b1
    H = relu(Z1)
    Z2 = H @ W2 + b2
    P = sigmoid(Z2)

    eps = 1e-12
    P_clip = np.clip(P, eps, 1 - eps)
    loss = -np.mean(y * np.log(P_clip) + (1 - y) * np.log(1 - P_clip))
    history.append(loss)

    delta2 = P - y
    dW2 = H.T @ delta2 / len(X)
    db2 = np.mean(delta2, axis=0)

    delta1 = (delta2 @ W2.T) * (Z1 > 0)
    dW1 = X.T @ delta1 / len(X)
    db1 = np.mean(delta1, axis=0)

    W2 -= eta * dW2
    b2 -= eta * db2
    W1 -= eta * dW1
    b1 -= eta * db1

print("final probabilities =")
print(sigmoid(relu(X @ W1 + b1) @ W2 + b2))
```

```
final probabilities =
[[3.78893539e-04]
 [9.99833072e-01]
 [9.99838096e-01]
 [1.27758015e-04]]
```

29.8 14.8 Training curve

```
plt.figure(figsize=(6, 4))
plt.plot(history)
plt.xlabel("iteration")
plt.ylabel("binary cross-entropy")
```

```
plt.title("Training a small neural network on XOR")
plt.show()
```



29.9 14.9 Exercises

1. Verify all shapes in the forward pass and backward pass.
2. Replace ReLU with sigmoid in the hidden layer and compare training behavior.
3. Increase the hidden dimension and observe what changes.
4. Explain why a linear model cannot solve XOR perfectly.
5. Derive the gradient formulas for the final layer.

29.10 14.10 Worked example

```
rng = np.random.default_rng(3)

X = np.array([
```

```
    [-1.0, -1.0],
    [-1.0,  1.0],
    [ 1.0, -1.0],
    [ 1.0,  1.0]
])

y = np.array([[0.0], [1.0], [1.0], [0.0]])

W1 = 0.3 * rng.normal(size=(2, 6))
b1 = np.zeros(6)
W2 = 0.3 * rng.normal(size=(6, 1))
b2 = np.zeros(1)

for _ in range(3000):
    Z1 = X @ W1 + b1
    H = relu(Z1)
    Z2 = H @ W2 + b2
    P = sigmoid(Z2)

    delta2 = P - y
    dW2 = H.T @ delta2 / len(X)
    db2 = np.mean(delta2, axis=0)
    delta1 = (delta2 @ W2.T) * (Z1 > 0)
    dW1 = X.T @ delta1 / len(X)
    db1 = np.mean(delta1, axis=0)

    W2 -= 0.4 * dW2
    b2 -= 0.4 * db2
    W1 -= 0.4 * dW1
    b1 -= 0.4 * db1

print((sigmoid(relu(X @ W1 + b1) @ W2 + b2) > 0.5).astype(int))
```

```
[[0]
 [1]
 [1]
 [0]]
```

30 15. Numerical Stability and Debugging

30.1 15.1 Why numerical stability matters

Even if a mathematical formula is correct, a direct implementation can fail because of floating-point limitations. In machine learning, common issues include:

- overflow in exponentials;
- underflow in probabilities;
- division by very small values;
- catastrophic cancellation;
- exploding or vanishing gradients;
- accidental shape mismatches.

30.2 15.2 Stable softmax

The naive softmax is

$$\mathrm{softmax}(x)_i = \frac{e^{x_i}}{\sum_j e^{x_j}}.$$

This can overflow when the entries of x are large.

A standard remedy is to subtract the maximum entry first.

```
def softmax_naive(x):
    ex = np.exp(x)
    return ex / np.sum(ex)

def softmax_stable(x):
    z = x - np.max(x)
    ez = np.exp(z)
    return ez / np.sum(ez)

x = np.array([1000.0, 1001.0, 1002.0])
```

```
print("stable softmax =", softmax_stable(x))
```

```
stable softmax = [0.09003057 0.24472847 0.66524096]
```

30.3 15.3 Stable logistic loss ideas

In practice, probabilities should often be clipped before taking logarithms.

```
p = np.array([1e-20, 0.2, 0.8, 1 - 1e-20])
p_clip = np.clip(p, 1e-12, 1 - 1e-12)

print("p =", p)
print("p_clip =", p_clip)
print("log(p_clip) =", np.log(p_clip))
```

```
p = [1.e-20 2.e-01 8.e-01 1.e+00]
p_clip = [1.e-12 2.e-01 8.e-01 1.e+00]
log(p_clip) = [-2.76310211e+01 -1.60943791e+00 -2.23143551e-01 -9.99977878e-
13]
```

30.4 15.4 Shape debugging

When code fails, shapes are often the culprit. Printing shapes systematically is an underrated debugging method.

```
X = np.random.randn(5, 3)
w = np.random.randn(3)
b = 0.5

print("X.shape =", X.shape)
print("w.shape =", w.shape)
print("X @ w shape =", (X @ w).shape)
print("(X @ w + b).shape =", (X @ w + b).shape)
```

```
X.shape = (5, 3)
w.shape = (3,)
X @ w shape = (5,)
(X @ w + b).shape = (5,)
```

30.5 15.5 Gradient checking revisited

Finite differences are a practical way to validate a backpropagation implementation on small examples.

```
def f_scalar(w):
    return np.sum(w**2)

def grad_scalar(w):
    return 2 * w

w = np.array([1.5, -2.0, 0.3])
g = grad_scalar(w)

h = 1e-6
g_fd = np.zeros_like(w)

for j in range(len(w)):
    e = np.zeros_like(w)
    e[j] = 1.0
    g_fd[j] = (f_scalar(w + h * e) - f_scalar(w - h * e)) / (2 * h)

print("analytic =", g)
print("finite difference =", g_fd)
```

```
analytic = [ 3.  -4.   0.6]
finite difference = [ 3.  -4.   0.6]
```

30.6 15.6 Common checklist

When a model behaves strangely, inspect the following.

1. Are the shapes what you think they are?
2. Are the dtypes appropriate?
3. Are the gradients exploding or vanishing?
4. Is the learning rate too large or too small?
5. Are you mixing row vectors and column vectors?
6. Are probabilities clipped before logarithms?
7. Are features on wildly different scales?

30.7 15.7 Exercises

1. Implement naive and stable softmax and compare them.
2. Create a deliberate shape error and fix it.
3. Run a finite-difference gradient test for a simple loss.
4. Explain why clipping probabilities avoids undefined logarithms.
5. Write your own debugging checklist for gradient-based optimization.

30.8 15.8 Worked example

```
x = np.array([500.0, 505.0, 510.0])

print("stable:", softmax_stable(x))

X = np.random.randn(7, 2)
w = np.random.randn(2, 1)

print("X.shape =", X.shape)
print("w.shape =", w.shape)
print("(X @ w).shape =", (X @ w).shape)
```

```
stable: [4.50940412e-05 6.69254912e-03 9.93262357e-01]
X.shape = (7, 2)
w.shape = (2, 1)
(X @ w).shape = (7, 1)
```

31 16. Capstone Mini-Projects

31.1 16.1 Project A: Linear regression on synthetic data

Tasks:

1. Generate a dataset with two input features and a linear target.
2. Fit the model using the normal equation.
3. Fit the model using gradient descent.
4. Compare the coefficient vectors.
5. Study the effect of noise and scaling.

Starter code:

```
rng = np.random.default_rng(100)

n = 250
X = rng.normal(size=(n, 2))
w_true = np.array([3.0, -1.5])
y = X @ w_true + 0.4 * rng.normal(size=n)

w_closed = np.linalg.solve(X.T @ X, X.T @ y)
print("w_closed =", w_closed)
```

```
w_closed = [ 2.96982269 -1.50309428]
```

31.2 16.2 Project B: Logistic regression on a tabular dataset

Tasks:

1. Load a CSV file with pandas.
2. Select numeric features and a binary target.
3. Handle missing values.
4. Convert to NumPy.
5. Train logistic regression.

6. Evaluate accuracy on a test split.

Starter code:

```
df = df_csv.copy()

if "target" in df.columns:
    feature_cols = [c for c in df.columns if c != "target" and pd.api.types.is_numeric_dtype
    X = df[feature_cols].to_numpy(dtype=float)
    y = df["target"].to_numpy(dtype=float)

    mu = X.mean(axis=0)
    sigma = X.std(axis=0)
    X = (X - mu) / sigma

    X = np.column_stack([np.ones(len(X)), X])

    rng = np.random.default_rng(99)
    perm = rng.permutation(len(X))
    split = int(0.8 * len(X))

    X_train = X[perm[:split]]
    X_test = X[perm[split:]]
    y_train = y[perm[:split]]
    y_test = y[perm[split:]]

    w = np.zeros(X.shape[1])

    for _ in range(1500):
        p = sigmoid(X_train @ w)
        grad = X_train.T @ (p - y_train) / len(X_train)
        w -= 0.1 * grad

    test_probs = sigmoid(X_test @ w)
    test_pred = (test_probs >= 0.5).astype(int)
    print("test accuracy =", np.mean(test_pred == y_test))
```

```
test accuracy = 0.4583333333333333
```

31.3 16.3 Project C: XOR with a neural network

Tasks:

1. Implement the forward pass.
2. Implement the backward pass.
3. Train the network on XOR.
4. Visualize the loss curve.
5. Experiment with different hidden sizes and activations.

Starter code:

```
X = np.array([
    [0.0, 0.0],
    [0.0, 1.0],
    [1.0, 0.0],
    [1.0, 1.0]
])

y = np.array([[0.0], [1.0], [1.0], [0.0]])
```

32 17. Summary and Outlook

32.1 17.1 Summary of the main ideas

The core message of this course is that machine learning can be understood very naturally through finite-dimensional linear algebra, calculus, and vectorized programming.

The main conceptual pillars were:

- arrays as concrete representations of vectors and matrices;
- slicing, masking, and broadcasting as tools for structured numerical work;
- matrix-vector products as the computational core of linear models;
- aggregation and statistics as the basis of losses and preprocessing;
- pandas as a labeled interface to tabular data;
- least squares as projection and optimization;
- logistic regression as a probability model with a convex loss;
- neural networks as compositions of affine maps and nonlinearities;
- numerical stability and shape awareness as practical necessities.

32.2 17.2 What students should now be able to do

A student who has worked through this script carefully should be able to:

1. load, inspect, and preprocess numerical tabular data;
2. represent machine learning models as matrix formulas;
3. derive gradients of standard losses;
4. implement linear regression and logistic regression from scratch;
5. implement a small neural network in pure NumPy;
6. debug shape mismatches and stability issues.

32.3 17.3 Natural continuations

Possible directions for a continuation course include:

- principal component analysis via singular value decomposition;

- stochastic gradient descent in greater depth;
- multiclass softmax regression;
- regularization paths and model selection;
- automatic differentiation;
- transition to PyTorch or JAX;
- optimization theory for machine learning;
- generalization bounds and statistical learning theory.

33 18. Suggested Homework Problems

1. Prove carefully that the least-squares residual is orthogonal to the column space of X.
2. Derive the logistic regression gradient in full detail.
3. Implement ridge regression and compare it to ordinary least squares on a polynomial task.
4. Create a notebook that illustrates the effect of feature scaling on gradient descent.
5. Build a two-layer network for XOR and verify the gradients numerically.
6. Write a short essay explaining the role of broadcasting in machine learning code.
7. Construct a dataset on which linear regression fails as a classifier and explain why logistic regression is better suited.
8. Show by experiment that different random seeds can influence optimization trajectories even if the final model class is the same.

34 19. Appendix: Reusable Utility Functions

```
def add_intercept(X):
    return np.column_stack([np.ones(len(X)), X])

def mse(y_true, y_pred):
    return np.mean((y_true - y_pred)**2)

def mae(y_true, y_pred):
    return np.mean(np.abs(y_true - y_pred))

def accuracy_score(y_true, y_pred):
    return np.mean(y_true == y_pred)

def standardize_train_test(X_train, X_test):
    mu = np.mean(X_train, axis=0)
    sigma = np.std(X_train, axis=0)
    sigma = np.where(sigma == 0, 1.0, sigma)
    return (X_train - mu) / sigma, (X_test - mu) / sigma, mu, sigma

def least_squares_fit(X, y):
    return np.linalg.solve(X.T @ X, X.T @ y)

def least_squares_predict(X, w):
    return X @ w

def logistic_predict_proba(X, w):
    return sigmoid(X @ w)

def logistic_predict(X, w, threshold=0.5):
    return (logistic_predict_proba(X, w) >= threshold).astype(int)

def logistic_fit_gd(X, y, eta=0.1, num_steps=1000):
    w = np.zeros(X.shape[1])
    history = []
    for _ in range(num_steps):
```

```
        p = sigmoid(X @ w)
        grad = X.T @ (p - y) / len(X)
        w -= eta * grad
        history.append(binary_cross_entropy(y, p))
    return w, history
```

34.1 19.1 Demonstration of the helper functions

```
rng = np.random.default_rng(123)

X = rng.normal(size=(80, 2))
w_true = np.array([1.5, -0.7])
y_reg = X @ w_true + 0.2 * rng.normal(size=80)

w_hat = least_squares_fit(X, y_reg)
print("least squares w_hat =", w_hat)

X_clf = add_intercept(X)
p_true = sigmoid(X_clf @ np.array([0.2, 1.0, -1.5]))
y_clf = (rng.random(80) < p_true).astype(float)

w_log, hist = logistic_fit_gd(X_clf, y_clf, eta=0.2, num_steps=800)
y_pred = logistic_predict(X_clf, w_log)

print("classification accuracy =", accuracy_score(y_clf, y_pred))
```

```
least squares w_hat = [ 1.48542799 -0.66725832]
classification accuracy = 0.775
```

35 20. Closing Remark

A useful way to summarize the philosophy of the course is this:

> **NumPy is not merely a programming library. It is a concrete language for expressing finite-dimensional mathematics.**

Once students see machine learning through that lens, many formulas stop looking mysterious. They become structured matrix computations, optimization problems, and geometric projections. That is precisely the bridge this script was designed to build.—

36 Kursziele

Dieser Kurs richtet sich an Mathematiker:innen und verfolgt zwei Ziele:

1. **NumPy als Sprache der numerischen linearen Algebra** verstehen.
2. **Grundideen des Machine Learning** mathematisch sauber formulieren und direkt in NumPy implementieren.

Am Ende sollen die Teilnehmenden einfache Modelle nicht nur anwenden, sondern selbst herleiten und programmieren können.

37 Voraussetzungen

- Grundlagen in Analysis und Linearer Algebra
- elementare Python-Kenntnisse
- keine Vorkenntnisse in Machine Learning nötig

38 Lernziele

Nach dem Kurs sollen die Teilnehmenden:

- Arrays, Broadcasting und Vektorisierung sicher verwenden,
- lineare und logistische Regression in NumPy implementieren,
- Gradientenverfahren verstehen und numerisch umsetzen,
- einfache neuronale Netze in Matrixform formulieren,
- numerische Stabilität und Shape-Probleme erkennen.

39 1. NumPy als mathematische Sprache

39.1 1.1 Arrays als Vektoren und Matrizen

In NumPy werden Vektoren und Matrizen als Arrays dargestellt.

```
import numpy as np

x = np.array([1.0, 2.0, 3.0])
A = np.array([[1.0, 0.0, 2.0],
              [0.0, 1.0, 1.0]])

print("x =", x)
print("A =\n", A)
print("shape von x:", x.shape)
print("shape von A:", A.ndim)
print("Dimension von A:", A.ndim)
```

```
x = [1. 2. 3.]
A =
 [[1. 0. 2.]
 [0. 1. 1.]]
shape von x: (3,)
shape von A: (2, 3)
Dimension von A: 2
```

39.2 1.2 Wichtige Grundbegriffe

- `shape`: Form des Arrays
- `ndim`: Anzahl der Achsen
- `dtype`: Datentyp
- Indexing und Slicing
- Views vs. Copies

```
B = np.arange(10)
print("B =", B)

print("B[2:6] =", B[2:6])

view = B[2:6]
view[0] = 99
print("Nach Änderung der View:")
print("view =", view)
print("B =", B)
```

```
B = [0 1 2 3 4 5 6 7 8 9]
B[2:6] = [2 3 4 5]
Nach Änderung der View:
view = [99  3  4  5]
B = [ 0  1 99  3  4  5  6  7  8  9]
```

> **!** Important
>
> **Didaktischer Hinweis**
> Gerade für mathematisch orientierte Studierende ist der Unterschied zwischen **View** und **Copy** wichtig:
> Nicht jede "Teilmenge" eines Arrays ist ein neues Objekt.

39.3 1.3 Normen und Skalarprodukt

```
x = np.array([1.0, -2.0, 2.0])
y = np.array([3.0, 1.0, -1.0])

l1 = np.sum(np.abs(x))
l2 = np.sqrt(np.sum(x**2))
dot = x @ y

print("||x||_1 =", l1)
print("||x||_2 =", l2)
print("<x,y> =", dot)
```

```
||x||_1 = 5.0
```

```
||x||_2 = 3.0
<x,y> = -1.0
```

39.4 Übung 1

1. Erzeuge einen Vektor $x \in \mathbb{R}^5$.
2. Berechne $\|x\|_1$, $\|x\|_2$ und $\|x\|_\infty$.
3. Implementiere das Skalarprodukt zweier Vektoren auf zwei Arten:
   - mit `@`
   - mit `np.sum(x * y)`

40 2. Vektorisierung und Broadcasting

40.1 2.1 Warum Vektorisierung?

In numerischer Mathematik und im Machine Learning vermeiden wir möglichst explizite Python-Schleifen.

```
x = np.linspace(0, 1, 5)
y = 3 * x + 2

print("x =", x)
print("y =", y)
```

```
x = [0.   0.25 0.5  0.75 1.  ]
y = [2.   2.75 3.5  4.25 5.  ]
```

40.2 2.2 Broadcasting

Broadcasting erlaubt Rechenoperationen zwischen Arrays unterschiedlicher, aber kompatibler Form.

```
X = np.array([[1.0, 2.0],
              [3.0, 4.0],
              [5.0, 6.0]])

b = np.array([10.0, 20.0])

print("X + b =\n", X + b)
```

```
X + b =
 [[11. 22.]
 [13. 24.]
 [15. 26.]]
```

Interpretation: Der Vektor b wird zeilenweise "mitgeführt".

40.3 2.3 Affine Abbildungen

Viele ML-Modelle beruhen auf der Form

$$y = Xw + b.$$

```
X = np.array([[1.0, 2.0],
              [0.0, 1.0],
              [3.0, 1.0]])
w = np.array([2.0, -1.0])
b = 0.5

y = X @ w + b
print("y =", y)
```

```
y = [ 0.5 -0.5  5.5]
```

40.4 Übung 2

1. Erzeuge eine Matrix $X \in \mathbb{R}^{4\times 3}$ und einen Vektor $w \in \mathbb{R}^3$.
2. Berechne Xw.
3. Addiere einen Bias-Term b.
4. Erkläre, warum Broadcasting hier funktioniert.

41 3. Lineare Algebra in NumPy

41.1 3.1 Matrix-Vektor- und Matrix-Matrix-Produkte

```
A = np.array([[1.0, 2.0],
              [3.0, 4.0]])
x = np.array([1.0, -1.0])
B = np.array([[2.0, 0.0],
              [1.0, 2.0]])

print("A @ x =", A @ x)
print("A @ B =\n", A @ B)
```

```
A @ x = [-1. -1.]
A @ B =
 [[ 4.  4.]
 [10.  8.]]
```

41.2 3.2 Transponieren

```
print("A^T =\n", A.T)
```

```
A^T =
 [[1. 3.]
 [2. 4.]]
```

41.3 3.3 Lineare Gleichungssysteme

Für $Ax = b$ sollte man **nicht** mit der Inversen arbeiten, sondern mit `np.linalg.solve`.

```
A = np.array([[3.0, 1.0],
              [1.0, 2.0]])
b = np.array([9.0, 8.0])

x = np.linalg.solve(A, b)
print("Lösung x =", x)
print("Prüfung A @ x =", A @ x)
```

```
Lösung x = [2. 3.]
Prüfung A @ x = [9. 8.]
```

41.4 3.4 Kondition und numerische Sensitivität

```
A = np.array([[1.0, 1.0],
              [1.0, 1.0001]])

cond = np.linalg.cond(A)
print("Konditionszahl =", cond)
```

```
Konditionszahl = 40002.00007491187
```

> **Note**
>
> Eine große Konditionszahl weist darauf hin, dass kleine Störungen in den Daten zu großen Änderungen in der Lösung führen können.

41.5 Übung 3

1. Löse ein lineares Gleichungssystem mit `np.linalg.solve`.
2. Vergleiche das Ergebnis mit einer Berechnung über `np.linalg.inv(A) @ b`.
3. Berechne die Konditionszahl einer Matrix deiner Wahl.

42 4. Statistik mit NumPy

42.1 4.1 Mittelwert, Varianz, Standardabweichung

```
data = np.array([[1.0, 2.0, 3.0],
                 [2.0, 3.0, 4.0],
                 [3.0, 4.0, 5.0]])

print("Mittelwert insgesamt:", np.mean(data))
print("Mittelwerte spaltenweise:", np.mean(data, axis=0))
print("Varianzen spaltenweise:", np.var(data, axis=0))
print("Standardabweichungen spaltenweise:", np.std(data, axis=0))
```

```
Mittelwert insgesamt: 3.0
Mittelwerte spaltenweise: [2. 3. 4.]
Varianzen spaltenweise: [0.66666667 0.66666667 0.66666667]
Standardabweichungen spaltenweise: [0.81649658 0.81649658 0.81649658]
```

42.2 4.2 Bedeutung von `axis`

- `axis=0`: spaltenweise Aggregation
- `axis=1`: zeilenweise Aggregation

```
print("Zeilensummen:", np.sum(data, axis=1))
print("Spaltensummen:", np.sum(data, axis=0))
```

```
Zeilensummen: [ 6.  9. 12.]
Spaltensummen: [ 6.  9. 12.]
```

42.3 4.3 Standardisierung

Ein klassischer Vorverarbeitungsschritt ist die Standardisierung:

$$\widetilde{X} = \frac{X - \mu}{\sigma}.$$

```
X = np.array([[1.0, 10.0],
              [2.0, 20.0],
              [3.0, 30.0],
              [4.0, 40.0]])

mu = np.mean(X, axis=0)
sigma = np.std(X, axis=0)

X_std = (X - mu) / sigma

print("mu =", mu)
print("sigma =", sigma)
print("standardisierte Daten =\n", X_std)
```

```
mu = [ 2.5 25. ]
sigma = [ 1.11803399 11.18033989]
standardisierte Daten =
 [[-1.34164079 -1.34164079]
 [-0.4472136  -0.4472136 ]
 [ 0.4472136   0.4472136 ]
 [ 1.34164079  1.34164079]]
```

42.4 Übung 4

1. Standardisiere einen kleinen Datensatz.
2. Überprüfe numerisch, dass Mittelwert ungefähr 0 und Standardabweichung ungefähr 1 sind.

43 5. Zufall und Reproduzierbarkeit

43.1 5.1 Zufallszahlen

```
rng = np.random.default_rng(42)

print("Zufallsvektor:", rng.random(5))
print("Normalverteilte Daten:", rng.standard_normal(5))
print("Zufallsindizes:", rng.integers(0, 10, size=5))
```

```
Zufallsvektor: [0.77395605 0.43887844 0.85859792 0.69736803 0.09417735]
Normalverteilte Daten: [-1.30217951  0.1278404  -0.31624259 -0.01680116 -
0.85304393]
Zufallsindizes: [5 3 1 9 7]
```

43.2 5.2 Reproduzierbarkeit

Für numerische Experimente im ML ist Reproduzierbarkeit zentral.

```
rng1 = np.random.default_rng(123)
rng2 = np.random.default_rng(123)

print(rng1.standard_normal(4))
print(rng2.standard_normal(4))
```

```
[-0.98912135 -0.36778665  1.28792526  0.19397442]
[-0.98912135 -0.36778665  1.28792526  0.19397442]
```

43.3 Übung 5

1. Erzeuge einen zufälligen Datensatz mit 100 Punkten in $\mathbb{R}^2$.
2. Erzeuge dazu ein zufälliges Label-Array mit Werten 0 und 1.
3. Teile die Daten in Trainings- und Testdaten auf.

44 Datenverarbeitung mit pandas

44.1 5.1 DataFrame als annotierte Matrix

```
import pandas as pd

df = pd.DataFrame({
    "age": [25, 30, 22, 40],
    "income": [50000, 60000, 45000, 80000],
    "target": [0, 1, 0, 1]
})

print(df)
```

```
   age  income  target
0   25   50000       0
1   30   60000       1
2   22   45000       0
3   40   80000       1
```

Interpretation: - Zeilen = Datenpunkte - Spalten = Features - entspricht einer Matrix X mit Zusatzstruktur

44.2 5.2 Auswahl und Projektion

```
print(df["age"])
print(df[["age", "income"]])
print(df.iloc[0])
```

```
0    25
1    30
```

```
2    22
3    40
Name: age, dtype: int64
   age  income
0   25   50000
1   30   60000
2   22   45000
3   40   80000
age          25
income    50000
target        0
Name: 0, dtype: int64
```

44.3 5.3 Filtern

```python
df_filtered = df[df["age"] > 25]
print(df_filtered)
```

```
   age  income  target
1   30   60000       1
3   40   80000       1
```

44.4 5.4 Fehlende Werte

```python
df.loc[2, "income"] = None

print(df.isna())
df_clean = df.dropna()
print(df_clean)
```

```
     age  income  target
0  False   False   False
1  False   False   False
2  False    True   False
3  False   False   False
   age   income  target
0   25  50000.0       0
```

```
1   30  60000.0       1
3   40  80000.0       1
```

44.5 5.5 Übergang zu NumPy

```python
X = df_clean[["age", "income"]].to_numpy()
y = df_clean["target"].to_numpy()

print("X =", X)
print("y =", y)
```

```
X = [[2.5e+01 5.0e+04]
 [3.0e+01 6.0e+04]
 [4.0e+01 8.0e+04]]
y = [0 1 1]
```

> **❗ Important**
>
> Ab diesem Punkt arbeiten wir wieder rein mathematisch mit NumPy.

44.6 Übung pandas

1. Lade einen Datensatz (CSV).
2. Entferne fehlende Werte.
3. Wähle zwei Features und ein Target.
4. Wandle alles in NumPy-Arrays um.

45 6. Lineare Regression

45.1 6.1 Problemstellung

Gegeben seien Daten (x_i, y_i), gesucht ist ein lineares Modell

$$\hat{y} = Xw.$$

Wir minimieren

$$\min_w \|Xw - y\|_2^2.$$

45.2 6.2 Normalengleichung

Die erste Optimalitätsbedingung führt auf

$$X^\top Xw = X^\top y.$$

```
X = np.array([[1.0, 1.0],
              [1.0, 2.0],
              [1.0, 3.0],
              [1.0, 4.0]])
y = np.array([2.0, 3.0, 5.0, 4.0])

w = np.linalg.solve(X.T @ X, X.T @ y)
print("Parameter w =", w)
```

```
Parameter w = [1.5 0.8]
```

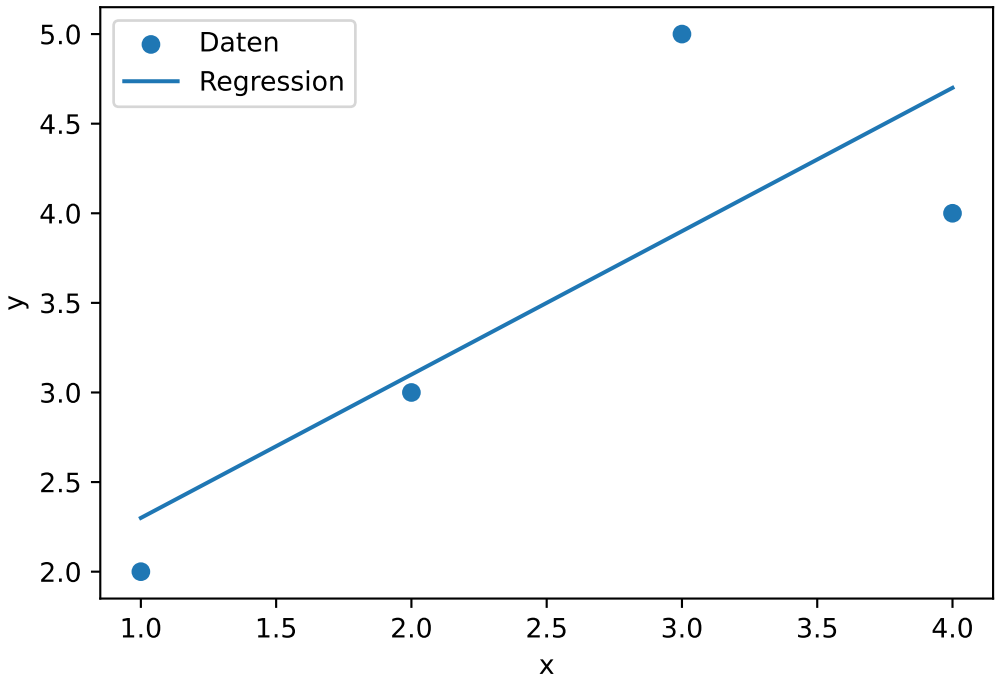
45.3 6.3 Vorhersage und Fehler

```
y_pred = X @ w
mse = np.mean((y_pred - y)**2)

print("Vorhersagen =", y_pred)
print("MSE =", mse)
```

```
Vorhersagen = [2.3 3.1 3.9 4.7]
MSE = 0.4500000000000002
```

45.4 6.4 Visualisierung

```
import matplotlib.pyplot as plt

x_plot = X[:, 1]

plt.figure(figsize=(6, 4))
plt.scatter(x_plot, y, label="Daten")
plt.plot(x_plot, y_pred, label="Regression")
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.show()
```



45.5 Übung 6

1. Implementiere lineare Regression über die Normalengleichung.
2. Füge Rauschen zu den Daten hinzu.
3. Untersuche, wie sich die Lösung verändert.

46 7. Gradientenverfahren

46.1 7.1 Gradient der quadratischen Verlustfunktion

Für

$$f(w) = \frac{1}{n}\|Xw - y\|_2^2$$

gilt

$$\nabla f(w) = \frac{2}{n}X^\top(Xw - y).$$

46.2 7.2 Gradient Descent

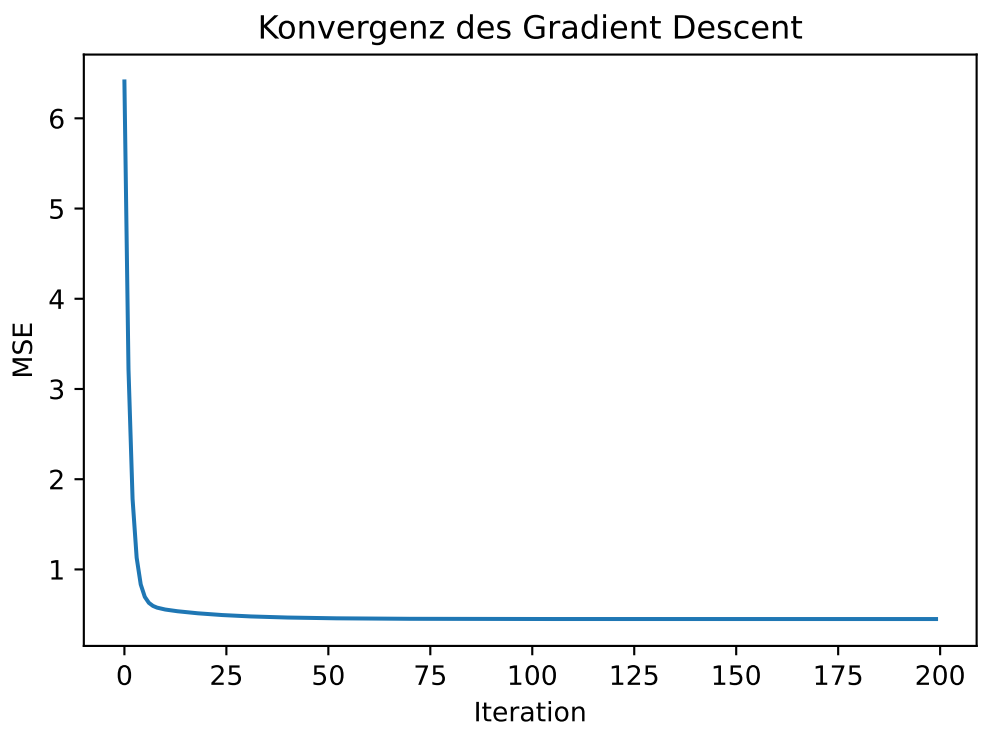
```
X = np.array([[1.0, 1.0],
              [1.0, 2.0],
              [1.0, 3.0],
              [1.0, 4.0]])
y = np.array([2.0, 3.0, 5.0, 4.0])

n, d = X.shape
w = np.zeros(d)
eta = 0.1

loss_history = []

for k in range(200):
    grad = (2 / n) * X.T @ (X @ w - y)
    w = w - eta * grad
    loss = np.mean((X @ w - y)**2)
    loss_history.append(loss)
```

```
print("Gefundene Parameter:", w)
print("Letzter Loss:", loss_history[-1])
```

```
Gefundene Parameter: [1.49747983 0.80085716]
Letzter Loss: 0.45000106073702084
```

46.3 7.3 Konvergenz visualisieren

```
plt.figure(figsize=(6, 4))
plt.plot(loss_history)
plt.xlabel("Iteration")
plt.ylabel("MSE")
plt.title("Konvergenz des Gradient Descent")
plt.show()
```



46.4 Übung 7

1. Experimentiere mit verschiedenen Lernraten `eta`.
2. Beobachte Konvergenz, langsames Lernen und Divergenz.
3. Vergleiche die GD-Lösung mit der Lösung aus der Normalengleichung.

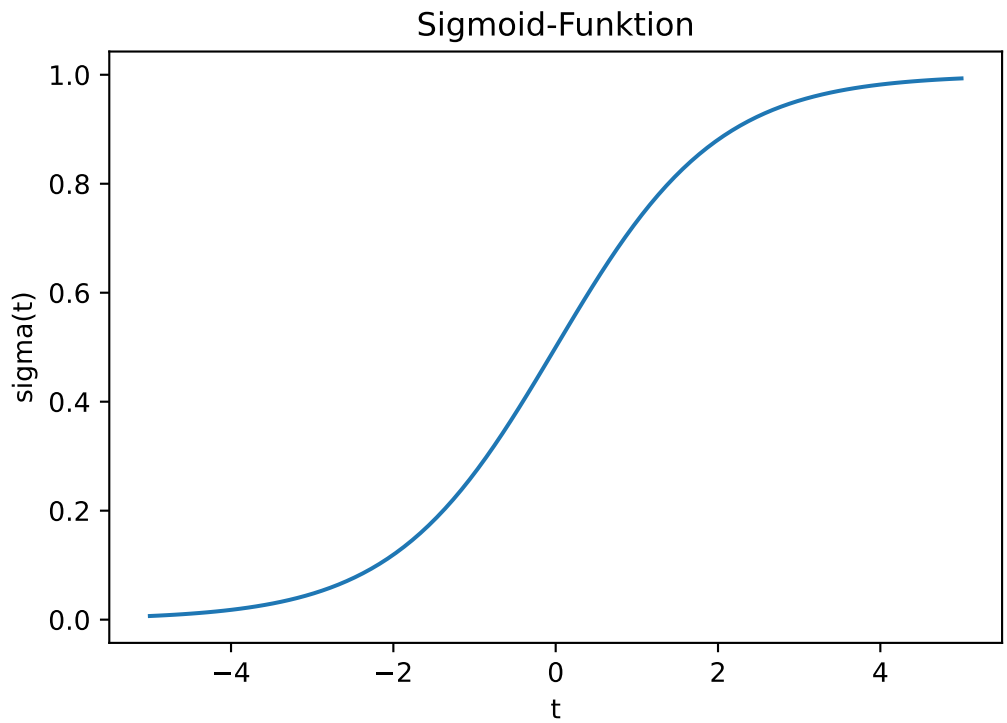
47 8. Logistische Regression

47.1 8.1 Binäre Klassifikation

Wir modellieren Wahrscheinlichkeiten mit der Sigmoid-Funktion

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

```
def sigmoid(t):
    return 1 / (1 + np.exp(-t))

t = np.linspace(-5, 5, 100)
plt.figure(figsize=(6, 4))
plt.plot(t, sigmoid(t))
plt.xlabel("t")
plt.ylabel("sigma(t)")
plt.title("Sigmoid-Funktion")
plt.show()
```



47.2 8.2 Modell

Für Datenmatrix X und Parameter w gilt

$$p(y = 1 \mid x) = \sigma(Xw).$$

47.3 8.3 Binary Cross Entropy

Die Verlustfunktion ist

$$L(w) = -\frac{1}{n} \sum_{i=1}^{n} \left(y_i \log p_i + (1 - y_i) \log(1 - p_i) \right).$$

```
def binary_cross_entropy(y_true, y_prob):
    eps = 1e-12
    y_prob = np.clip(y_prob, eps, 1 - eps)
    return -np.mean(y_true * np.log(y_prob) + (1 - y_true) * np.log(1 - y_prob))
```

47.4 8.4 Training per Gradient Descent

```
X_raw = np.array([[0.0, 0.0],
                  [0.0, 1.0],
                  [1.0, 0.0],
                  [1.0, 1.0],
                  [1.5, 1.0],
                  [2.0, 1.0]])

y = np.array([0, 0, 0, 1, 1, 1], dtype=float)

X = np.column_stack([np.ones(len(X_raw)), X_raw])

n, d = X.shape
w = np.zeros(d)
eta = 0.5
history = []

for _ in range(1000):
    p = sigmoid(X @ w)
    grad = (X.T @ (p - y)) / n
    w -= eta * grad
    history.append(binary_cross_entropy(y, p))

print("w =", w)
print("finaler Loss =", history[-1])
```

```
w = [-8.43141241  5.56359354  5.49551652]
finaler Loss = 0.03031078985439453
```

47.5 8.5 Vorhersagen

```
p = sigmoid(X @ w)
y_pred = (p >= 0.5).astype(int)

print("Wahrscheinlichkeiten:", p)
print("Vorhergesagte Klassen:", y_pred)
print("Trefferquote:", np.mean(y_pred == y))
```

```
Wahrscheinlichkeiten: [2.17866013e-04 5.04073600e-02 5.37675122e-02 9.32623019e-
01
 9.95546025e-01 9.99723021e-01]
Vorhergesagte Klassen: [0 0 0 1 1 1]
Trefferquote: 1.0
```

47.6 Übung 8

1. Implementiere logistische Regression selbst.
2. Variiere die Lernrate und die Anzahl der Iterationen.
3. Ersetze die Trainingsdaten durch einen eigenen 2D-Datensatz.

48 9. Shapes, Masking und Datenvorverarbeitung

48.1 9.1 Reshape

```
x = np.arange(12)
X = x.reshape(3, 4)

print("x =", x)
print("X =\n", X)
```

```
x = [ 0  1  2  3  4  5  6  7  8  9 10 11]
X =
 [[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
```

48.2 9.2 Neue Achsen

```
x = np.array([1.0, 2.0, 3.0])

print("x.shape =", x.shape)
print("x[:, None].shape =", x[:, None].shape)
print("x[None, :].shape =", x[None, :].shape)
```

```
x.shape = (3,)
x[:, None].shape = (3, 1)
x[None, :].shape = (1, 3)
```

48.3 9.3 Boolean Masking

```
X = np.array([[1.0, 10.0],
              [-1.0, 20.0],
              [2.0, 30.0],
              [-2.0, 40.0]])

mask = X[:, 0] > 0
print("Maske:", mask)
print("Gefilterte Daten =\n", X[mask])
```

```
Maske: [ True False  True False]
Gefilterte Daten =
 [[ 1. 10.]
 [ 2. 30.]]
```

48.4 Übung 9

1. Erzeuge einen Datensatz mit positiven und negativen Werten.
2. Filtere alle Zeilen heraus, deren erste Komponente negativ ist.
3. Erläutere die Rolle von Masking im Preprocessing.

49 10. Einfaches neuronales Netz

49.1 10.1 Modellidee

Ein einfaches Netzwerk mit einer versteckten Schicht kann als

$$f(x) = W_2\,\phi(XW_1 + b_1) + b_2$$

geschrieben werden.

Wir verwenden als Aktivierungsfunktion die ReLU

$$\mathrm{ReLU}(t) = \max(0, t).$$

49.2 10.2 Forward Pass

```
def relu(x):
    return np.maximum(0, x)

rng = np.random.default_rng(0)

X = np.array([[0.0, 0.0],
              [0.0, 1.0],
              [1.0, 0.0],
              [1.0, 1.0]])

W1 = rng.standard_normal((2, 4))
b1 = np.zeros(4)
W2 = rng.standard_normal((4, 1))
b2 = np.zeros(1)

H = relu(X @ W1 + b1)
Y = H @ W2 + b2
```

```
print("Hidden layer H =\n", H)
print("Output Y =\n", Y)
```

```
Hidden layer H =
 [[0.         0.         0.         0.        ]
 [0.         0.36159505 1.30400005 0.94708096]
 [0.12573022 0.         0.64042265 0.10490012]
 [0.         0.22949019 1.9444227  1.05198108]]
Output Y =
 [[ 0.        ]
 [-1.23118103]
 [-0.48330477]
 [-1.45883668]]
```

49.3 10.3 Bemerkung zur Backpropagation

Für einen vollständigen Kurs sollte hier die Kettenregel in Matrixform entwickelt werden. In einer ersten Einführung genügt oft:

1. Vorwärtsdurchlauf definieren,
2. Verlustfunktion formulieren,
3. Gradienten der einzelnen Parameter herleiten,
4. Update-Regel implementieren.

> **Tip**
>
> Gerade für Mathematiker:innen lohnt es sich, Backpropagation als wiederholte Anwendung der Kettenregel auf affine Abbildungen und nichtlineare Aktivierungen zu präsentieren.

49.4 Übung 10

1. Implementiere ein kleines neuronales Netz mit einer versteckten Schicht.
2. Verwende ReLU als Aktivierung.
3. Erzeuge einen kleinen künstlichen Datensatz und berechne den Forward Pass.

50 11. Numerische Stabilität

50.1 11.1 Problem: Overflow bei `exp`

```
x = np.array([1000.0, 1001.0, 1002.0])
```

Die naive Softmax-Implementierung

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

kann numerisch instabil sein.

50.2 11.2 Stabile Softmax

```
def softmax_stable(x):
    z = x - np.max(x)
    exp_z = np.exp(z)
    return exp_z / np.sum(exp_z)

print("stabile Softmax:", softmax_stable(x))
```

```
stabile Softmax: [0.09003057 0.24472847 0.66524096]
```

50.3 Übung 11

1. Implementiere eine naive Softmax.
2. Vergleiche sie mit der stabilen Version.
3. Diskutiere das Problem des Overflows.

51 12. Mini-Projekte

51.1 Projekt A: Lineare Regression

- Simuliere Daten
- fitte ein Modell
- interpretiere die Parameter
- visualisiere Residuen

51.2 Projekt B: Logistische Regression

- konstruiere einen binären Datensatz
- trainiere ein Modell
- berechne Accuracy und Loss

51.3 Projekt C: NumPy-only Neuronales Netz

- ein Hidden Layer
- ReLU oder Sigmoid
- Training mit Gradient Descent
- Diskussion von Overfitting und Lernrate

52 13. Zusammenfassung

Die zentralen NumPy-Ideen für Machine Learning sind:

- **Arrays als Träger mathematischer Objekte**
- **Vektorisierung statt Schleifen**
- **Matrixrechnung für Modelle**
- **statistische Aggregation für Datenanalyse**
- **Gradientenverfahren für Optimierung**
- **Shape-Kompetenz für sauberen Code**
- **numerische Stabilität für robuste Implementierungen**

53 14. Weiterführende Themen

Für einen Anschlusskurs bieten sich an:

- Regularisierung (Ridge, Lasso)
- PCA mit SVD
- stochastischer Gradient Descent
- Mini-Batching
- automatische Differentiation
- Übergang zu PyTorch oder JAX

54 Hausaufgabenideen

1. Implementiere lineare Regression mit und ohne Normalengleichung.
2. Implementiere logistische Regression für einen selbst erzeugten Datensatz.
3. Untersuche den Einfluss der Lernrate auf die Konvergenz.
4. Erstelle eine stabile Softmax-Funktion.
5. Dokumentiere typische Shape-Fehler und ihre Korrektur.

55 Literaturhinweise

- NumPy-Dokumentation
- G. Strang: *Linear Algebra and Learning from Data*
- Hastie, Tibshirani, Friedman: *The Elements of Statistical Learning*
- Goodfellow, Bengio, Courville: *Deep Learning*